*Physics 172*

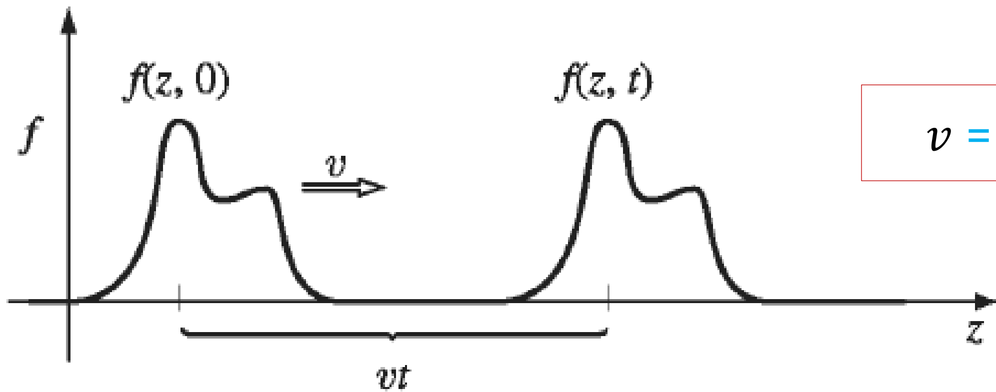
# *Chapter 9*
# *Electromagnetic Waves*

Griffiths, D. *Introduction to Electrodynamics*.

# 9.1 Waves in One Dimension

## 9.1.1 The Wave Equation

Mechanical wave in a stretched string:




$v$ = *wave speed*

- For a single pulse:

$$g(z) \equiv f(z,0) \quad \text{is initial shape}$$

- The pulse shape does not change:

$$f(z,t) = f(z - vt, 0) = g(z - vt)$$

### 9.1.1 The Wave Equation

Mechanical wave in a stretched string:

$$f(z,t) = g(z - vt) = g(u)$$

- Taking derivatives:

$$\frac{\partial f}{\partial t} = (-v)\frac{dg}{du} \implies \frac{\partial^2 f}{\partial t^2} = v^2 \frac{d^2 g}{du^2}$$

$$\frac{\partial f}{\partial z} = \frac{dg}{du} \implies \frac{\partial^2 f}{\partial z^2} = \frac{d^2 g}{du^2}$$

- Combining:

$$\frac{\partial^2 f}{\partial t^2} = v^2 \frac{\partial^2 f}{\partial z^2}$$

*The wave equation!*

### 9.1.1 The Wave Equation

For a stretched string with tension *F* and linear density $\mu$ :

$$\frac{\partial^2 f}{\partial t^2} = \left(\frac{F}{\mu}\right)\frac{\partial^2 f}{\partial z^2} \quad \text{with} \quad v = \pm\sqrt{\left(\frac{F}{\mu}\right)}$$

- General solution:

$$f(z,t) = g(z - vt) + h(z + vt)$$

*Valid wave functions*

$$f_1(z,t) = Ae^{-b(z-vt)^2}, \quad f_2(z,t) = A\sin[b(z-vt)], \quad f_3(z,t) = \frac{A}{b(z-vt)^2+1}$$

*Not valid wave functions*

$$f_4(z,t) = Ae^{-b(bz^2+vt)}, \quad \text{and} \quad f_5(z,t) = A\sin(bz)\cos(bvt)^3$$

## 9.1.2 Sinusoidal Waves

Harmonic waves:

- At $t = 0$, the *waveform* is:

$$f(z,0) = A\cos(kz+\delta) = A\cos\left(\frac{2\pi}{\lambda}z+\delta\right) \text{ where } k = \frac{2\pi}{\lambda}$$

$\lambda$ = *wavelength*

- For a travelling wave:

$$f(z,t) = A\cos[k(z-vt)+\delta] = A\cos(kz-\omega t+\delta)$$

$$\text{where } \omega = kv = 2\pi\nu = \frac{2\pi}{T}$$

$\nu$ = *frequency*

$T$ = *period*

## 9.1.2 Sinusoidal Waves

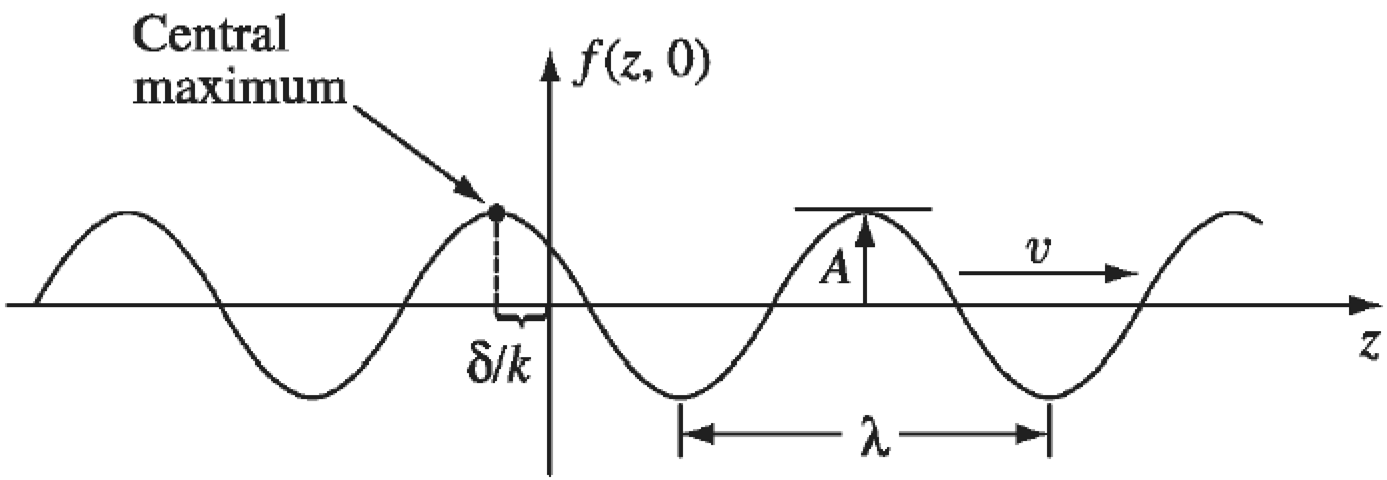
Harmonic waves:

- Travelling wave to $+z$:

$$f(z,t) = A\cos(kz - \omega t + \delta)$$

- Travelling wave to $-z$:

$$f(z,t) = A\cos(kz + \omega t - \delta)$$
$$= A\cos(-kz - \omega t + \delta)$$



## 9.1.2 Sinusoidal Waves

Complex notation:

- Travelling wave to $+z$:

$$f(z,t) = A\cos(kz - \omega t + \delta) = \mathrm{Re}\left[Ae^{-i(kz-\omega t+\delta)}\right]$$

- The complex wave function:

$$\tilde{f}(z,t) = \tilde{A}e^{i(kz-\omega t)}$$

$$\tilde{A} = Ae^{i\delta}$$ *Complex amplitude*

$$f(z,t) = \mathrm{Re}\left[\tilde{f}(z,t)\right]$$ *Real function*

## 9.1.2 Sinusoidal Waves

Superposition of waves:

- For simultaneous waves in a medium:

$$f(z,t) = \sum f_j(z,t) = \sum A_j \cos\left(k_j z - \omega_j t + \delta_j\right)$$

- The complex wave function:

$$\tilde{f}(z,t) = \sum \tilde{A}_j e^{i\left(k_j z - \omega_j t\right)}$$

- Fourier synthesis:

$$\tilde{f}(z,t) = \int_{-\infty}^{\infty} \tilde{A}(k) e^{i(kz - \omega t)} dk$$

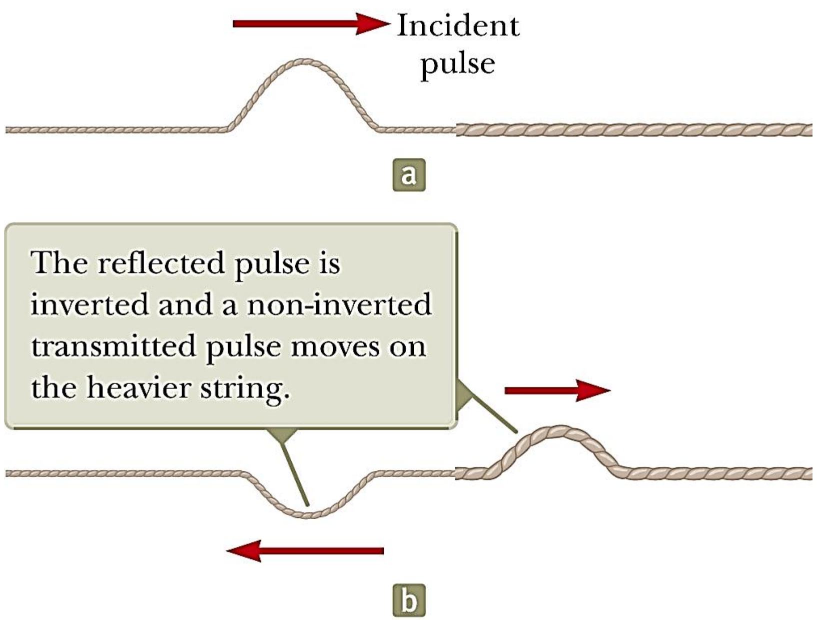
$\omega = \omega(k)$ *= function of k*

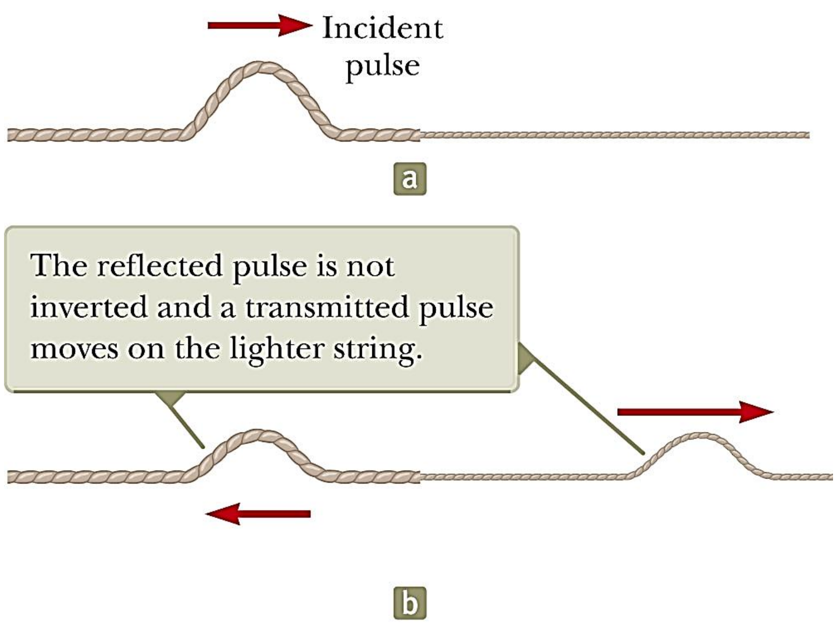
*Any waveform can be represented as a linear superposition of sinusoidal waves!*

### 9.1.3 Boundary Conditions

Reflection and transmission:

- When a light string is attached to a heavier string.

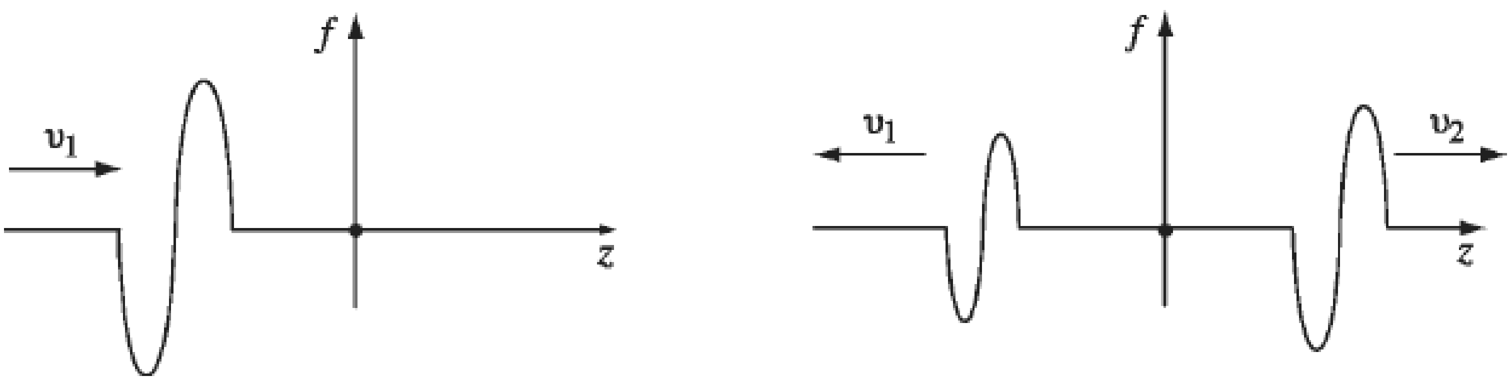


### 9.1.3 Boundary Conditions

Reflection and transmission:

- Conservation of energy governs the waves.
  - When a pulse is broken up into reflected and transmitted parts at a boundary, the sum of the energies of the two pulses must equal the energy of the original pulse.
  - When a pulse travels from medium *1* to medium *2* and $v_1 > v_2$, it is *inverted* upon reflection. (Medium *1* is denser than *2*.)
  - When a pulse travels from medium *1* to medium *2* and $v_1 < v_2$, it is *not inverted* upon reflection. (Medium *1* is denser than *2*.)



(a) Incident pulse (b) Reflected and transmitted pulses

## 9.1.3 Boundary Conditions

Reflection and transmission:

- Incident wave:

$$\tilde{f}_I(z,t) = \tilde{A}_I e^{i(k_1 z - \omega t)} \quad (z < 0)$$

- Reflected wave:

$$\tilde{f}_R(z,t) = \tilde{A}_R e^{i(-k_1 z - \omega t)} \quad (z < 0)$$

- Transmitted wave:

$$\tilde{f}_T(z,t) = \tilde{A}_T e^{i(k_2 z - \omega t)} \quad (z > 0)$$

(All parts of the system oscillate at the same frequency.)

$$\frac{\lambda_1}{\lambda_2} = \frac{k_2}{k_1} = \frac{v_1}{v_2}$$

### 9.1.3 Boundary Conditions

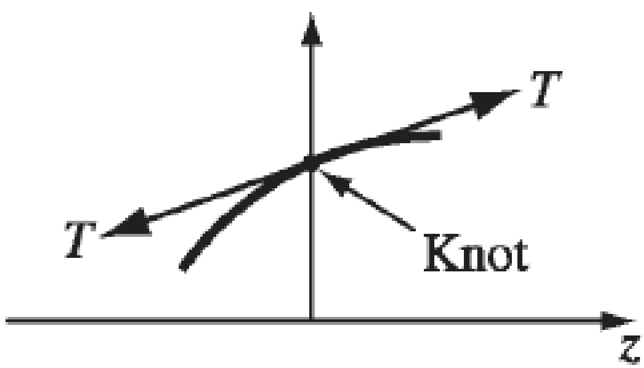
- The complete wave function:

$$\tilde{f}(z,t) = \begin{cases} \tilde{A}_I e^{i(k_1 z - \omega t)} + \tilde{A}_R e^{i(-k_1 z - \omega t)} & (z < 0) \\ \tilde{A}_T e^{i(k_2 z - \omega t)} & (z > 0) \end{cases}$$

- Continuity at the boundary ( $z = 0$ ):

$$\tilde{f}(0^-,t) = \tilde{f}(0^+,t) \quad \text{and} \quad \left.\frac{\partial y}{\partial x}\right|_{0^-} = \left.\frac{\partial y}{\partial x}\right|_{0^+}$$

*Slope is continuous*



### 9.1.3 Boundary Conditions

- At $z = 0$ :

$$\begin{cases} \tilde{A}_I + \tilde{A}_R = \tilde{A}_T \\ k_1(\tilde{A}_I - \tilde{A}_R) = k_2 \tilde{A}_T \end{cases}$$

- Solution :

$$\begin{cases} \tilde{A}_R = \left(\dfrac{k_1 - k_2}{k_1 + k_2}\right) \tilde{A}_I \\ \tilde{A}_T = \left(\dfrac{2k_1}{k_1 + k_2}\right) \tilde{A}_I \end{cases}$$

- In terms of the velocities :

$$\tilde{A}_R = \left(\frac{v_2 - v_1}{v_1 + v_2}\right) \tilde{A}_I \qquad \tilde{A}_T = \left(\frac{2v_2}{v_1 + v_2}\right) \tilde{A}_I$$

### 9.1.3 Boundary Conditions

- The real amplitudes and phases :

$$A_R e^{i\delta_R} = \left(\frac{v_2 - v_1}{v_1 + v_2}\right) A_I e^{i\delta_I} \qquad A_T e^{i\delta_T} = \left(\frac{2v_2}{v_1 + v_2}\right) A_I e^{i\delta_I}$$

- When $v_2 > v_1$ : $\delta_R = \delta_I$ ; $\delta_T = \delta_I$

$$A_R = \left(\frac{v_2 - v_1}{v_1 + v_2}\right) A_I \qquad A_T = \left(\frac{2v_2}{v_1 + v_2}\right) A_I$$

- When $v_2 < v_1$ : $\delta_R = \delta_I$ ; $\delta_T = \delta_I$

$$A_R e^{i\delta_R} = -\left(\frac{v_2 - v_1}{v_1 + v_2}\right) A_I e^{i\delta_I} = \left(\frac{v_2 - v_1}{v_1 + v_2}\right) A_I e^{i\delta_I} e^{-i\pi} \qquad \delta_R = \delta_I - \pi$$

($180°$ *phase shift* )

$$A_R = -\left(\frac{v_2 - v_1}{v_1 + v_2}\right) A_I \qquad A_T = \left(\frac{2v_2}{v_1 + v_2}\right) A_I$$

## 9.1.3 Boundary Conditions

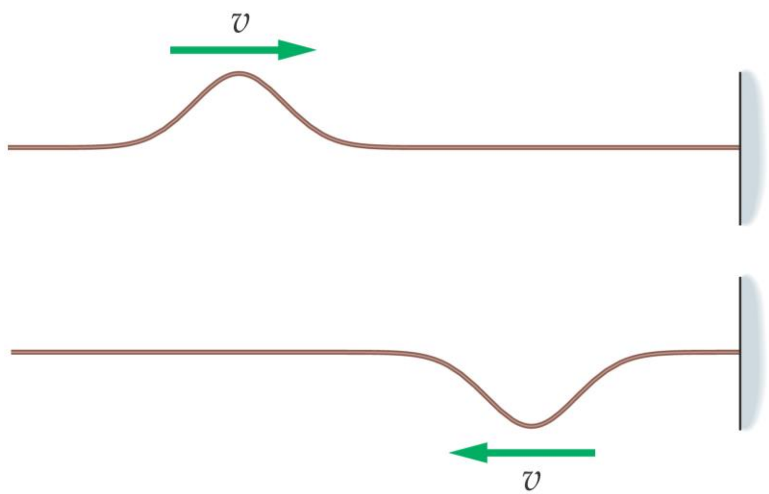
Example: Special cases

- When $v_2 = 0$ (fixed end)

$$\delta_R = \delta_I - \pi$$

$$A_R = -A_I \qquad A_T = 0$$

Reflected wave is “upside down”

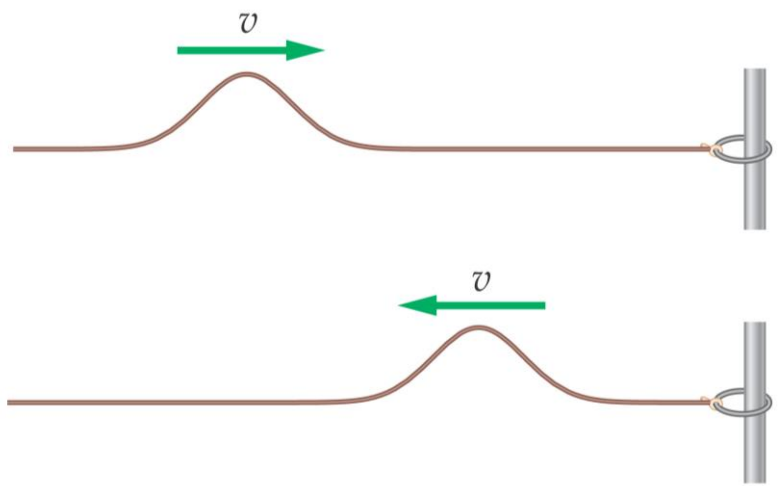



- When second medium is absent (“free” end)

$$\left.\frac{\partial y}{\partial x}\right|_{0^-} = \left.\frac{\partial y}{\partial x}\right|_{0^+} = 0$$

$$A_R = A_I \qquad A_T = 0$$

Wave is reflected without inversion.



### 9.1.4 Polarization

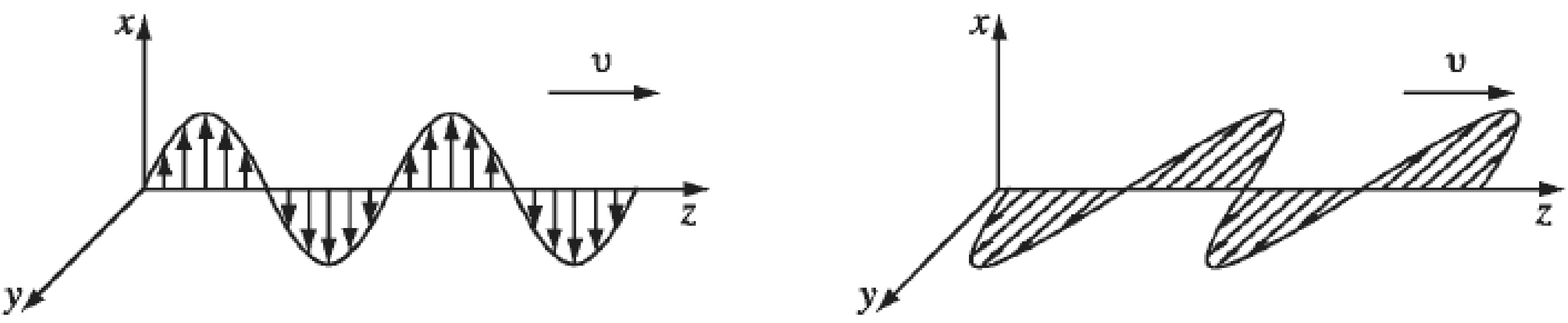
For transverse waves

- Wave function written as a vector

$$\tilde{\boldsymbol{f}}(z,t) = \tilde{A}_x e^{i(k_1 z - \omega t)}\hat{\mathbf{x}} + \tilde{A}_y e^{i(-k_1 z - \omega t)}\hat{\mathbf{y}}$$

$$= \tilde{A} e^{i(k_1 z - \omega t)}\hat{\mathbf{n}}$$

$\hat{\mathbf{n}}$ = Polarization vector




(a) Vertical polarization

(b) Horizontal polarization

## 9.2 Electromagnetic Waves in Vacuum

### 9.2.1 Wave Equation for $\mathbf{E}$ and $\mathbf{B}$

Maxwell's equations for vacuum ( $\rho = 0$; $I = 0$ )

$$\left.\begin{array}{llll} \text{(i)} & \nabla \cdot \mathbf{E} = 0, & \text{(iii)} & \nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t}, \\[2ex] \text{(ii)} & \nabla \cdot \mathbf{B} = 0, & \text{(iv)} & \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \dfrac{\partial \mathbf{E}}{\partial t}. \end{array}\right\} \qquad (9.40)$$

- Decoupling the equations

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = \nabla \times \left(-\frac{\partial \mathbf{B}}{\partial t}\right)$$

$$= -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}) = -\mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2},$$

*Same with* **B**

### 9.2.1 Wave Equation for $\mathbf{E}$ and $\mathbf{B}$

- Since $\boldsymbol{\nabla} \cdot \mathbf{E} = 0$ , $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$

$$\nabla^2 \mathbf{E} = \epsilon_0 \mu_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \text{and} \quad \nabla^2 \mathbf{B} = \epsilon_0 \mu_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

- For 1-dimensional wave

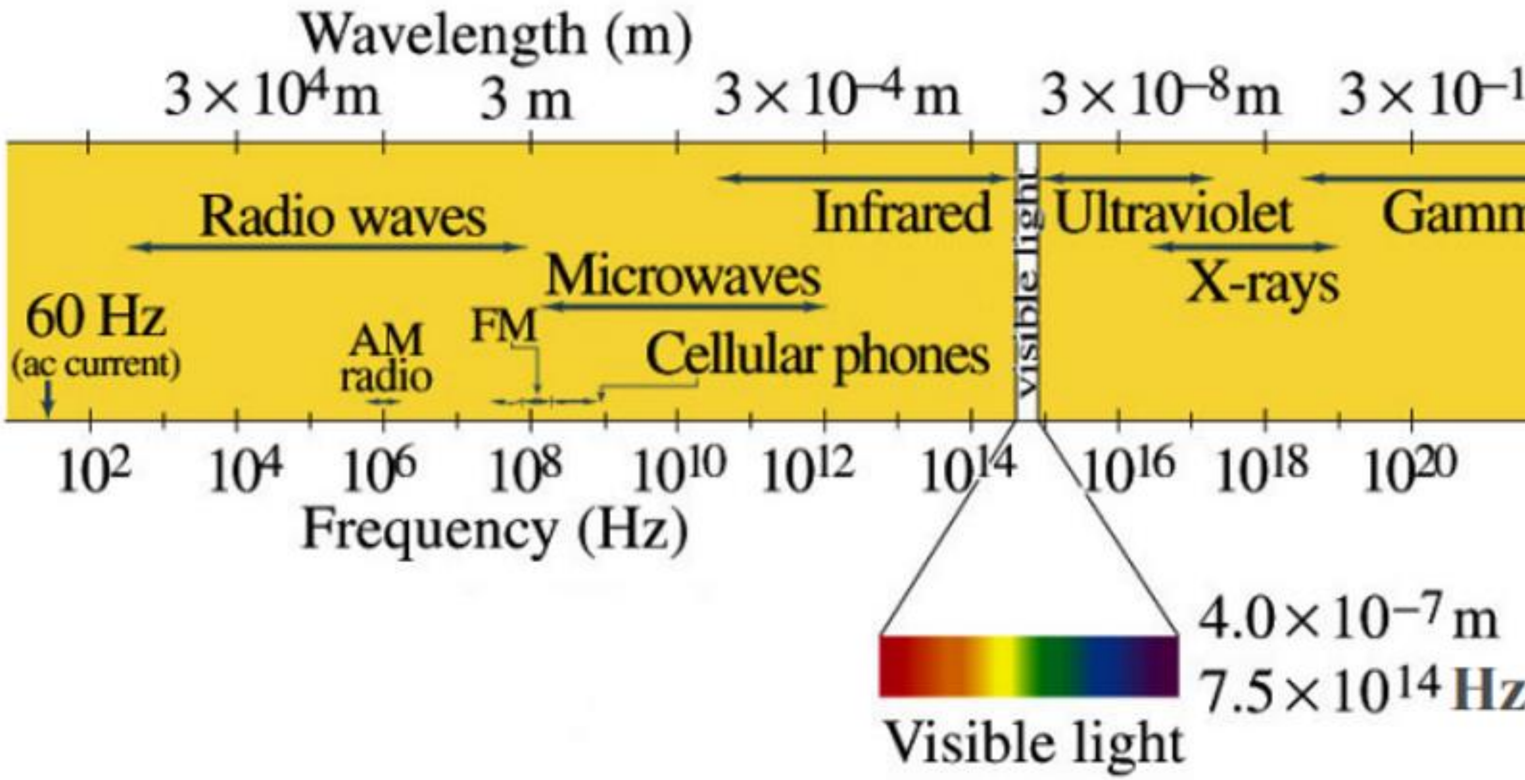
$$\frac{\partial^2 \mathbf{E}}{\partial z^2} = \epsilon_0 \mu_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

- Identify

$$\frac{1}{v^2} = \epsilon_0 \mu_0 \quad \text{or} \quad v = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = c = 3 \times 10^8 \text{ m/s}$$

### 9.2.1 Wave Equation for $\mathbf{E}$ and $\mathbf{B}$

- The electromagnetic spectrum



## 9.2.2 Monochromatic Plane Waves

The wave function depends on *z* only [ $\mathbf{E} = \mathbf{E}(z, t)$ ]

- The complex wave functions

$$\tilde{\mathbf{E}}(z,t) = \tilde{\mathbf{E}}_0 e^{i(kz-\omega t)}, \quad \tilde{\mathbf{B}}(z,t) = \tilde{\mathbf{B}}_0 e^{i(kz-\omega t)} \tag{9.43}$$

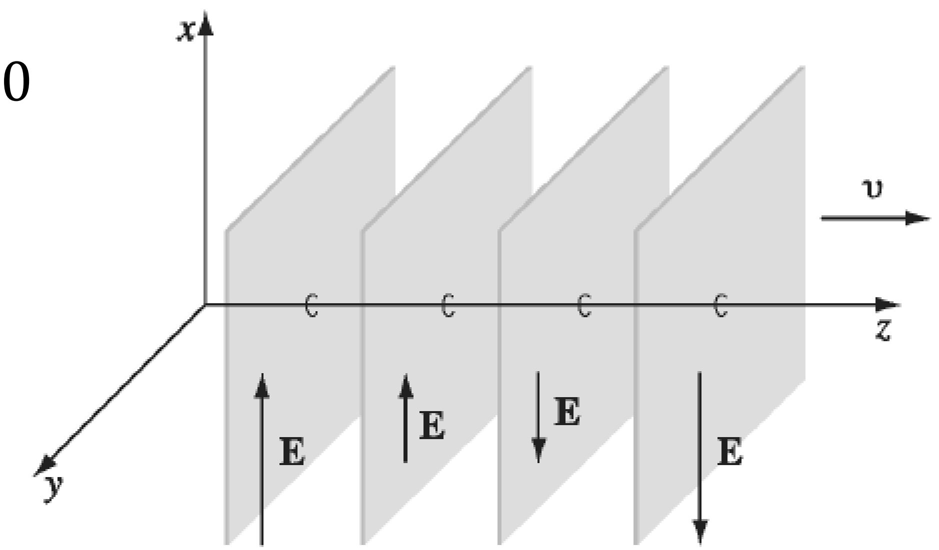
- Apply $\nabla \cdot \tilde{\mathbf{E}} = 0$ , $\nabla \cdot \tilde{\mathbf{B}} = 0$

$$\frac{\partial(\tilde{\mathbf{E}})_z}{\partial z} = 0 \ , \qquad \frac{\partial(\tilde{\mathbf{B}})_z}{\partial z} = 0$$

$$(\tilde{\mathbf{E}}_0)_z = 0 \ , \qquad (\tilde{\mathbf{B}}_0)_z = 0$$

*No static fields.*

*Electromagnetic waves are transverse waves.*



## 9.2.2 Monochromatic Plane Waves

The wave function depends on *z* only [ $\mathbf{E} = \mathbf{E}(z, t)$ ]

- Apply $\nabla \times \tilde{\mathbf{E}} = -\partial\tilde{\mathbf{B}}/\partial t$

$$-k(\tilde{E}_0)_y = \omega(\tilde{B}_0)_x, \quad k(\tilde{E}_0)_x = \omega(\tilde{B}_0)_y.$$

$$\tilde{\mathbf{B}}_0 = \frac{k}{\omega}(\hat{\mathbf{z}} \times \tilde{\mathbf{E}}_0)$$

- **E** and **B** are *in phase* and *mutually perpendicular.*

$$B_0 = \frac{k}{\omega}E_0 = \frac{1}{c}E_0 \tag{9.47}$$

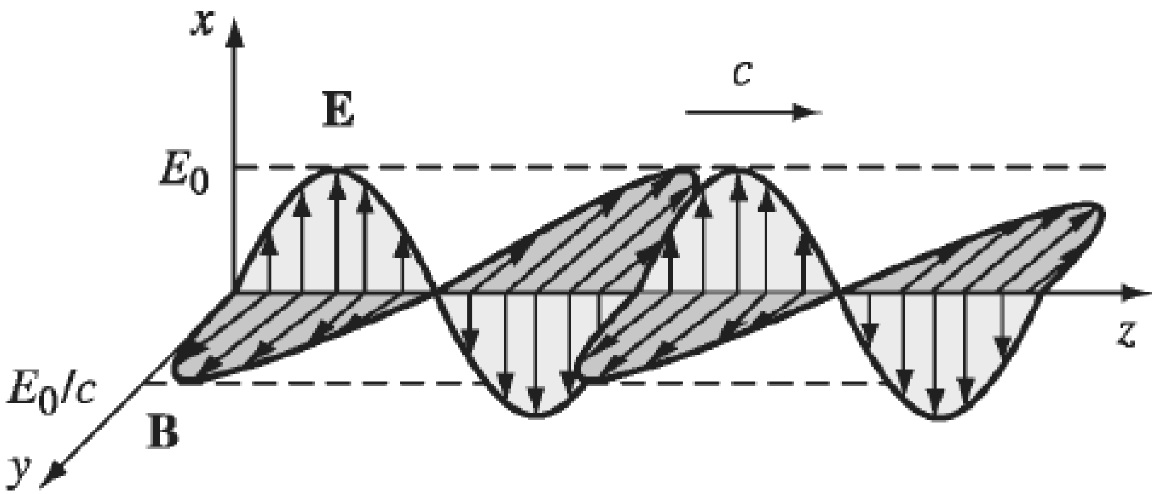
### 9.2.2 Monochromatic Plane Waves

Example:

$$\tilde{\mathbf{E}}(z,t) = \tilde{E}_0 e^{i(kz-\omega t)}\,\hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}(z,t) = \frac{1}{c}\tilde{E}_0 e^{i(kz-\omega t)}\,\hat{\mathbf{y}}.$$

- The real fields:

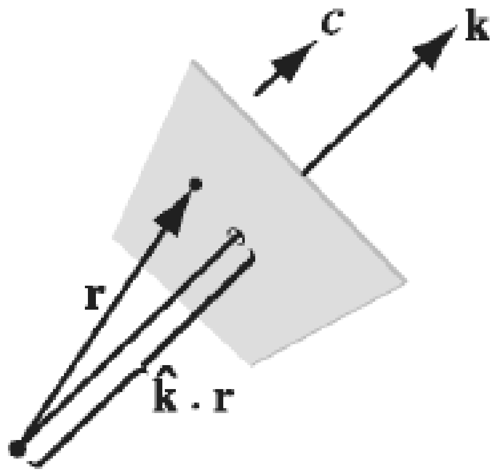
$$\mathbf{E}(z,t) = E_0\cos(kz-\omega t+\delta)\,\hat{\mathbf{x}}, \quad \mathbf{B}(z,t) = \frac{1}{c}E_0\cos(kz-\omega t+\delta)\,\hat{\mathbf{y}}.$$



## 9.2.2 Monochromatic Plane Waves

For waves travelling in some arbitrary direction:

- Define the propagation vector **k**: $kz \Longrightarrow \mathbf{k} \cdot \mathbf{r}$




$$\tilde{\mathbf{E}}(\mathbf{r}, t) = \tilde{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\, \hat{\mathbf{n}},$$

$$\tilde{\mathbf{B}}(\mathbf{r}, t) = \frac{1}{c}\tilde{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}(\hat{\mathbf{k}} \times \hat{\mathbf{n}}) = \frac{1}{c}\hat{\mathbf{k}} \times \tilde{\mathbf{E}},$$

$\hat{\mathbf{n}}$ = Polarization vector

- The real fields:

$$\mathbf{E}(\mathbf{r}, t) = E_0 \cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta)\, \hat{\mathbf{n}} \tag{9.51}$$

$$\mathbf{B}(\mathbf{r}, t) = \frac{1}{c} E_0 \cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta)\, (\hat{\mathbf{k}} \times \hat{\mathbf{n}}) \tag{9.52}$$

### 9.2.3 Energy and Momentum in Electromagnetic Waves

- Energy density:

$$u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) = \epsilon_0 E^2$$

$$u = \epsilon_0 {E_0}^2 \cos^2(kz - \omega t + \delta)$$

- Energy flux density:

$$\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B})$$

$$\mathbf{S} = c\epsilon_0 {E_0}^2 \cos^2(kz - \omega t + \delta)\,\hat{\mathbf{z}} = cu\,\hat{\mathbf{z}}$$

- Momentum density:

$$\mathbf{g} = \frac{1}{c^2}\mathbf{S}$$

$$\mathbf{g} = \frac{1}{c}\epsilon_0 {E_0}^2 \cos^2(kz - \omega t + \delta)\,\hat{\mathbf{z}} = \frac{1}{c} u\,\hat{\mathbf{z}} \qquad (9.59)$$

### 9.2.3 Energy and Momentum in Electromagnetic Waves

Time averaged values:

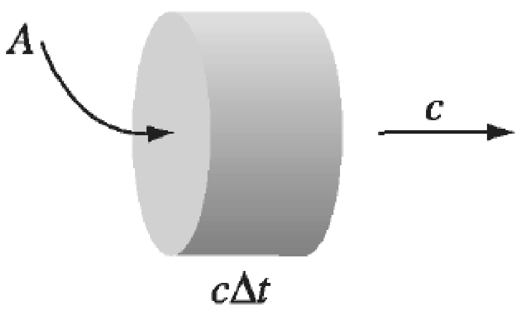
$$\langle u \rangle = \frac{1}{2}\epsilon_0 {E_0}^2 \quad \langle \mathbf{S} \rangle = \frac{1}{2}c\epsilon_0 {E_0}^2 \hat{\mathbf{z}} \quad \langle \mathbf{g} \rangle = \frac{1}{2c}\epsilon_0 {E_0}^2 \hat{\mathbf{z}}$$

- The wave intensity:

$$I = \langle \boldsymbol{S} \rangle = \frac{1}{2}c\epsilon_0 {E_0}^2$$

- Radiation pressure: ( $\Delta \boldsymbol{p} = \langle \mathbf{g} \rangle Ac\Delta t$ )

$$P = \frac{1}{A}\frac{\Delta \boldsymbol{p}}{\Delta t} = c\langle \mathbf{g} \rangle = \frac{1}{2}\epsilon_0 {E_0}^2 = \frac{I}{c}$$



## 9.3 Electromagnetic Waves in Matter

### 9.3.1 Propagation in Linear Media

Inside matter, but no free charges and currents:

$$\left.\begin{array}{llll} \text{(i)} & \nabla\cdot\mathbf{D}=0, & \text{(iii)} & \nabla\times\mathbf{E}=-\dfrac{\partial\mathbf{B}}{\partial t}, \\ \text{(ii)} & \nabla\cdot\mathbf{B}=0, & \text{(iv)} & \nabla\times\mathbf{H}=\dfrac{\partial\mathbf{D}}{\partial t}. \end{array}\right\}$$

- For linear, homogeneous media: ( $\mathbf{D}=\epsilon\mathbf{E}$ , $\mathbf{B}=\mu\mathbf{H}$ )

$$\left.\begin{array}{llll} \text{(i)} & \nabla\cdot\mathbf{E}=0, & \text{(iii)} & \nabla\times\mathbf{E}=-\dfrac{\partial\mathbf{B}}{\partial t}, \\ \text{(ii)} & \nabla\cdot\mathbf{B}=0, & \text{(iv)} & \nabla\times\mathbf{B}=\mu\epsilon\dfrac{\partial\mathbf{E}}{\partial t}, \end{array}\right\} \qquad (9.67)$$

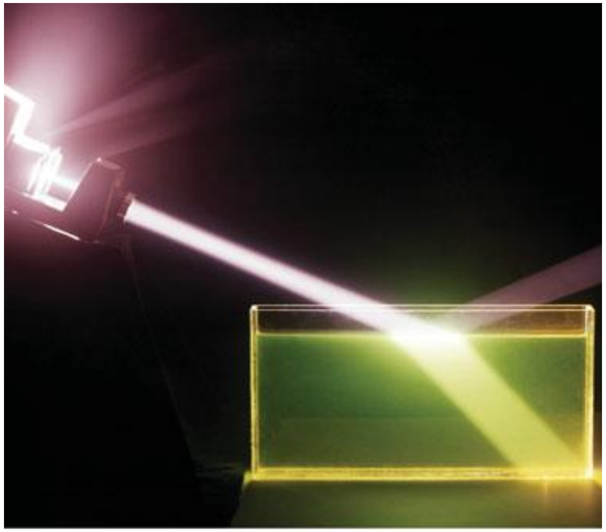
- Identify

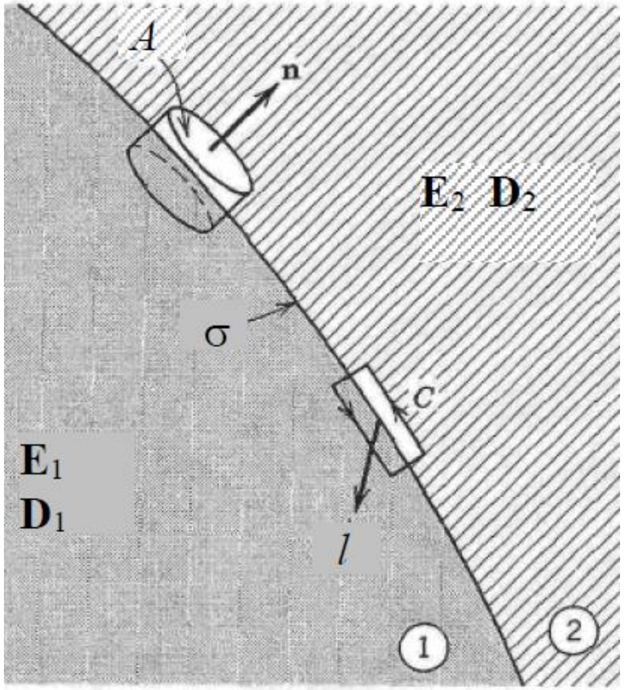
$$v=\frac{1}{\sqrt{\epsilon\mu}}=\frac{c}{n}$$

$n$ = *index of refraction*

### 9.3.1 Propagation in Linear Media

When a wave passes from one transparent medium into another :



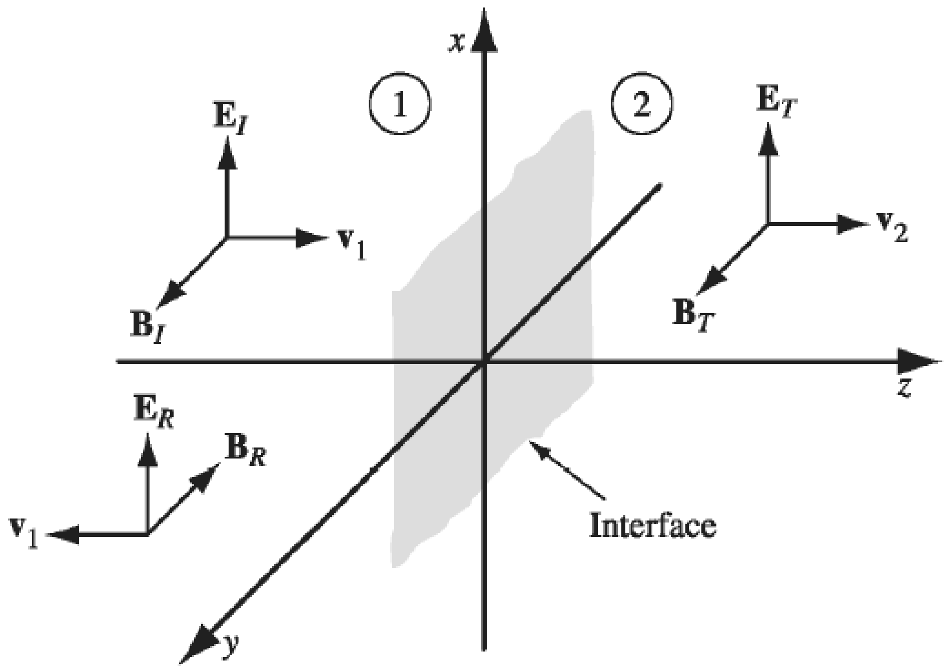



- Boundary conditions:

$$\left.\begin{array}{llll}\text{(i)} & \epsilon_1 E_1^{\perp} = \epsilon_2 E_2^{\perp}, & \text{(iii)} & \mathbf{E}_1^{\parallel} = \mathbf{E}_2^{\parallel}, \\ \text{(ii)} & B_1^{\perp} = B_2^{\perp}, & \text{(iv)} & \dfrac{1}{\mu_1}\mathbf{B}_1^{\parallel} = \dfrac{1}{\mu_2}\mathbf{B}_2^{\parallel}. \end{array}\right\} \quad (9.74)$$

## 9.3.2 Reflection and Transmission at Normal Incidence

At the plane interface :

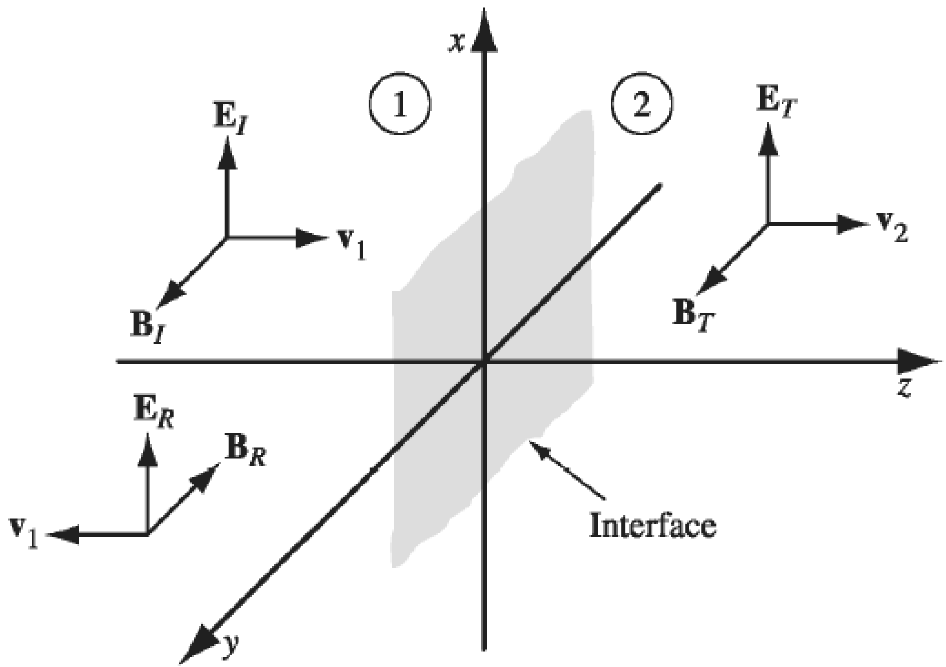



- Incident wave:

$$\left.\begin{aligned} \tilde{\mathbf{E}}_I(z,t) &= \tilde{E}_{0_I} e^{i(k_1 z - \omega t)}\,\hat{\mathbf{x}}, \\ \tilde{\mathbf{B}}_I(z,t) &= \frac{1}{v_1}\tilde{E}_{0_I} e^{i(k_1 z - \omega t)}\,\hat{\mathbf{y}}. \end{aligned}\right\}$$

## 9.3.2 Reflection and Transmission at Normal Incidence

At the plane interface :




- Reflected wave:

$$\left.\begin{aligned}\tilde{\mathbf{E}}_R(z,t) &= \tilde{E}_{0_R} e^{i(-k_1 z-\omega t)}\,\hat{\mathbf{x}},\\ \tilde{\mathbf{B}}_R(z,t) &= -\frac{1}{v_1}\tilde{E}_{0_R} e^{i(-k_1 z-\omega t)}\,\hat{\mathbf{y}},\end{aligned}\right\}$$

- Transmitted wave:

$$\left.\begin{aligned}\tilde{\mathbf{E}}_T(z,t) &= \tilde{E}_{0_T} e^{i(k_2 z-\omega t)}\,\hat{\mathbf{x}},\\ \tilde{\mathbf{B}}_T(z,t) &= \frac{1}{v_2}\tilde{E}_{0_T} e^{i(k_2 z-\omega t)}\,\hat{\mathbf{y}},\end{aligned}\right\}$$

## 9.3.2 Reflection and Transmission at Normal Incidence

At the plane interface ($z = 0$) :

- From (iii): $\mathbf{E}_1^{\parallel} = \mathbf{E}_2^{\parallel}$

$$\tilde{E}_{0I} + \tilde{E}_{0R} = \tilde{E}_{0T}$$

- From (iv): $\mathbf{B}_1^{\parallel}/\mu_1 = \mathbf{B}_2^{\parallel}/\mu_2$

$$\left(\tilde{E}_{0I} - \tilde{E}_{0R}\right)/\mu_1 v_1 = \left(\tilde{E}_{0T}\right)/\mu_2 v_2$$

$$\tilde{E}_{0I} - \tilde{E}_{0R} = \beta \tilde{E}_{0T} \quad \text{where} \quad \beta = \mu_1 v_1/\mu_2 v_2$$

- Solving:

$$\begin{cases} \tilde{E}_{0I} + \tilde{E}_{0R} = \tilde{E}_{0T} \\ \tilde{E}_{0I} - \tilde{E}_{0R} = \beta \tilde{E}_{0T} \end{cases} \Longrightarrow \begin{cases} \tilde{E}_{0R} = \left(\dfrac{1-\beta}{1+\beta}\right) \tilde{E}_{0I} \\ \tilde{E}_{0T} = \left(\dfrac{2}{1+\beta}\right) \tilde{E}_{0I} \end{cases}$$

## 9.3.2 Reflection and Transmission at Normal Incidence

At the plane interface :

- In terms of the index of refraction

$$\tilde{E}_{0R} = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)\tilde{E}_{0I} \qquad \tilde{E}_{0T} = \left(\frac{2n_1}{n_1 + n_2}\right)\tilde{E}_{0I}$$

- The intensities: $I = v\epsilon {E_0}^2/2$

$$\frac{I_R}{I_I} = \left(\frac{\tilde{E}_{0R}}{\tilde{E}_{0I}}\right)^2 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 \equiv R$$

*Reflection coefficient*

$$\frac{I_T}{I_I} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1}\left(\frac{\tilde{E}_{0T}}{\tilde{E}_{0I}}\right)^2 = \frac{4n_1 n_2}{(n_1 + n_2)^2} \equiv T$$

*Transmission coefficient*

### 9.3.3 Reflection and Transmission at Oblique Incidence

At the plane interface :

$$\tilde{\mathbf{E}}_R(\mathbf{r}, t) = \tilde{\mathbf{E}}_{0_R} e^{i(\mathbf{k}_R \cdot \mathbf{r} - \omega t)}$$

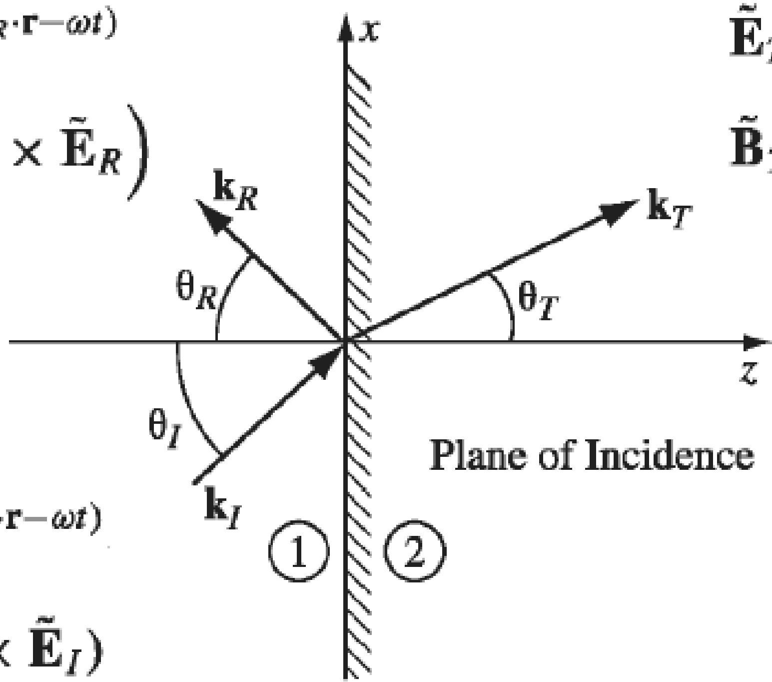
$$\tilde{\mathbf{B}}_R(\mathbf{r}, t) = \frac{1}{v_1}\left(\hat{\mathbf{k}}_R \times \tilde{\mathbf{E}}_R\right)$$

*(Reflected)*

$$\tilde{\mathbf{E}}_T(\mathbf{r}, t) = \tilde{\mathbf{E}}_{0_T} e^{i(\mathbf{k}_T \cdot \mathbf{r} - \omega t)}$$

$$\tilde{\mathbf{B}}_T(\mathbf{r}, t) = \frac{1}{v_2}\left(\hat{\mathbf{k}}_T \times \tilde{\mathbf{E}}_T\right)$$

*(Transmitted)*




*(Incident)*

$$\tilde{\mathbf{E}}_I(\mathbf{r}, t) = \tilde{\mathbf{E}}_{0_I} e^{i(\mathbf{k}_I \cdot \mathbf{r} - \omega t)}$$

$$\tilde{\mathbf{B}}_I(\mathbf{r}, t) = \frac{1}{v_1}(\hat{\mathbf{k}}_I \times \tilde{\mathbf{E}}_I)$$

- With the wavenumbers:

$$k_I v_1 = k_R v_1 = k_T v_2 = \omega \quad \text{or} \quad k_I = k_R = \left(\frac{v_2}{v_1}\right) k_T$$

## 9.3.3 Reflection and Transmission at Oblique Incidence

Apply boundary conditions ( at $z = 0$) :

- From (i): $\epsilon_1 \mathbf{E}_1^{\perp} = \epsilon_2 \mathbf{E}_2^{\perp}$

$$\epsilon_1 (\tilde{\boldsymbol{E}}_{0I} e^{i\mathbf{k}_I \cdot \mathbf{r}} + \tilde{\boldsymbol{E}}_{0R} e^{i\mathbf{k}_R \cdot \mathbf{r}})_z = \epsilon_2 (\tilde{\boldsymbol{E}}_{0T} e^{i\mathbf{k}_T \cdot \mathbf{r}})_z$$

  - The exponential factors must cancel:

$$\mathbf{k}_I \cdot \mathbf{r} = \mathbf{k}_R \cdot \mathbf{r} = \mathbf{k}_T \cdot \mathbf{r} \qquad (9.94)$$

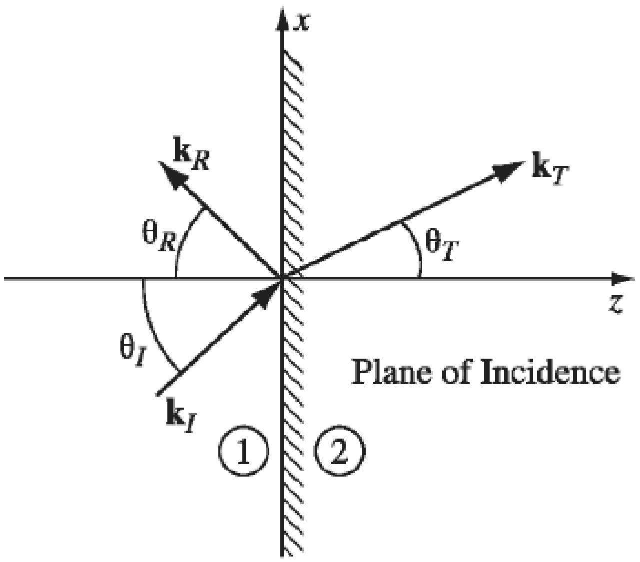
1. $k_I \sin\theta_I = k_R \sin\theta_R \quad \Longrightarrow \quad \theta_R = \theta_I$ *Law of reflection*
2. $k_I \sin\theta_I = k_T \sin\theta_T$

$$\Longrightarrow \quad \frac{\sin\theta_T}{\sin\theta_I} = \frac{v_2}{v_1} = \frac{n_1}{n_2}$$ *Snell's law*

### 9.3.3 Reflection and Transmission at Oblique Incidence

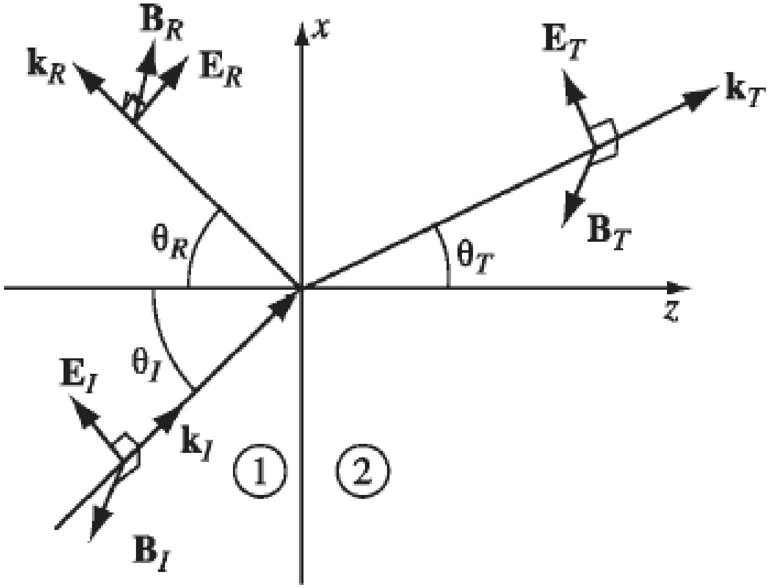
Apply boundary conditions ( at $z = 0$) :

$$\left.\begin{aligned}
&\text{(i)}\quad \epsilon_1\left(\tilde{\mathbf{E}}_{0_I}+\tilde{\mathbf{E}}_{0_R}\right)_z=\epsilon_2\left(\tilde{\mathbf{E}}_{0_T}\right)_z\\
&\text{(ii)}\quad \left(\tilde{\mathbf{B}}_{0_I}+\tilde{\mathbf{B}}_{0_R}\right)_z=\left(\tilde{\mathbf{B}}_{0_T}\right)_z\\
&\text{(iii)}\quad \left(\tilde{\mathbf{E}}_{0_I}+\tilde{\mathbf{E}}_{0_R}\right)_{x,y}=\left(\tilde{\mathbf{E}}_{0_T}\right)_{x,y}\\
&\text{(iv)}\quad \frac{1}{\mu_1}\left(\tilde{\mathbf{B}}_{0_I}+\tilde{\mathbf{B}}_{0_R}\right)_{x,y}=\frac{1}{\mu_2}\left(\tilde{\mathbf{B}}_{0_T}\right)_{x,y}
\end{aligned}\right\}$$



### 9.3.3 Reflection and Transmission at Oblique Incidence

When incident wave is polarized parallel to plane of incidence:




- From (i):

$$\epsilon_1\left(-\tilde{E}_{0I}\sin\theta_I + \tilde{E}_{0R}\sin\theta_R\right)$$
$$= \epsilon_2(-\tilde{E}_{0T}\sin\theta_T)$$

- From (iii):

$$\tilde{E}_{0I}\cos\theta_I + \tilde{E}_{0R}\cos\theta_R = \tilde{E}_{0T}\cos\theta_T$$

- From (iv):

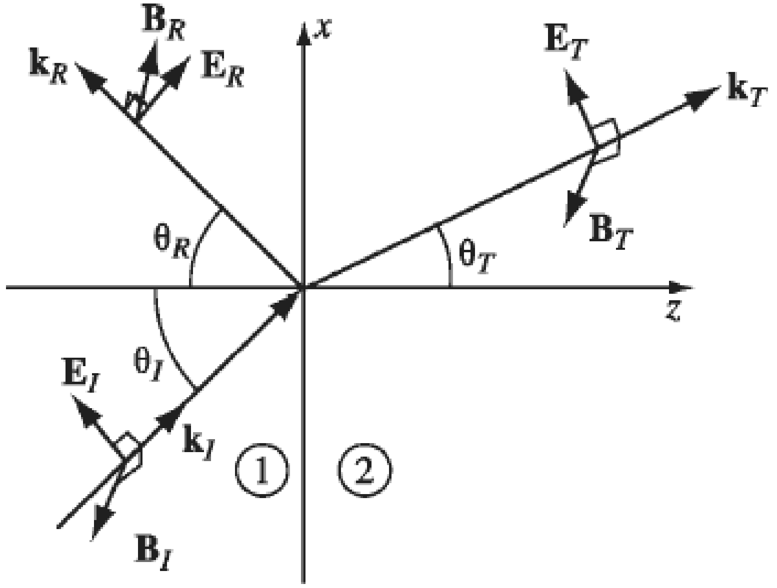
$$(\tilde{E}_{0I} - \tilde{E}_{0R})/\mu_1 v_1 = (\tilde{E}_{0T})/\mu_2 v_2$$

### 9.3.3 Reflection and Transmission at Oblique Incidence

When incident wave is polarized parallel to plane of incidence:

- Solution(iv):

$$\begin{cases} \tilde{E}_{0I}+\tilde{E}_{0R}=\dfrac{\cos\theta_T}{\cos\theta_I}\tilde{E}_{0T}=\alpha\tilde{E}_{0T} \\ \tilde{E}_{0I}-\tilde{E}_{0R}=\dfrac{\mu_1 v_1}{\mu_2 v_2}\tilde{E}_{0T}=\beta\tilde{E}_{0T} \end{cases}$$




$$\Longrightarrow \begin{cases} \tilde{E}_{0R}=\left(\dfrac{\alpha-\beta}{\alpha+\beta}\right)\tilde{E}_{0I} \\ \tilde{E}_{0T}=\left(\dfrac{2}{\alpha+\beta}\right)\tilde{E}_{0I} \end{cases}$$

*Fresnel's equations*

$$\alpha=\frac{\cos\theta_T}{\cos\theta_I}\ ;\ \ \beta=\frac{\mu_1 v_1}{\mu_2 v_2}=\frac{\mu_1 n_2}{\mu_2 n_1}$$

### 9.3.3 Reflection and Transmission at Oblique Incidence

When incident wave is polarized parallel to plane of incidence:

- Fresnel equations:

$$\tilde{E}_{0R} = \left(\frac{\alpha - \beta}{\alpha + \beta}\right)\tilde{E}_{0I}\ ; \qquad \tilde{E}_{0T} = \left(\frac{2}{\alpha + \beta}\right)\tilde{E}_{0I} \qquad (9.109)$$

- The intensities: $I = \frac{1}{2} v \epsilon {E_0}^2 cos\theta$

$$\frac{I_R}{I_I} = \left(\frac{\tilde{E}_{0R}}{\tilde{E}_{0I}}\right)^2 = \left(\frac{\alpha - \beta}{\alpha + \beta}\right)^2 \equiv R$$

*Reflection coefficient*

$$\frac{I_T}{I_I} = \frac{\epsilon_2 v_2}{\epsilon_1 v_1}\left(\frac{\tilde{E}_{0T}}{\tilde{E}_{0I}}\right)^2 \frac{\cos\theta_T}{\cos\theta_I} = \frac{4\alpha\beta}{(\alpha + \beta)^2} \equiv T$$

*Transmission coefficient*

### 9.3.3 Reflection and Transmission at Oblique Incidence

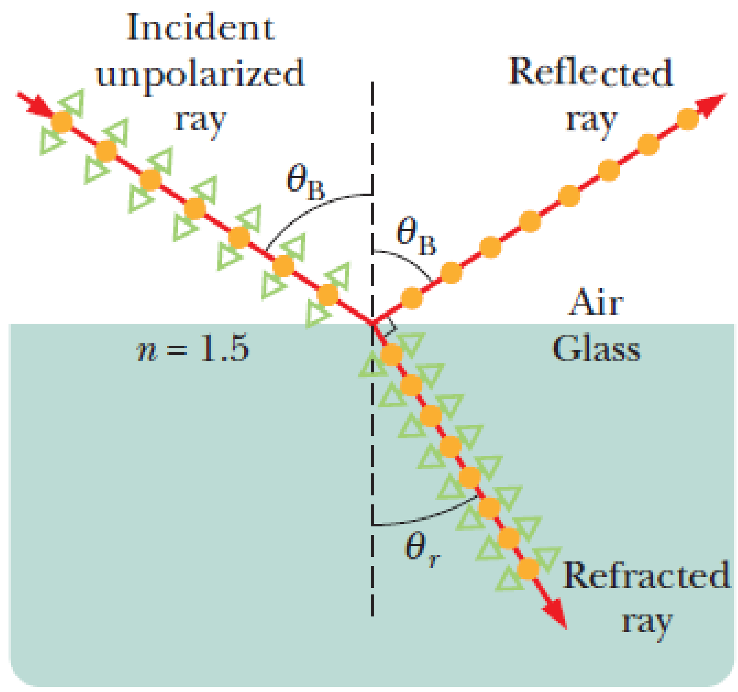
When incident wave is polarized parallel to plane of incidence:

- Brewster's angle:

$$\tilde{E}_{0R} = 0 \quad \text{when} \quad \alpha = \beta$$

$$\theta_I = \theta_B$$

$$\tan \theta_B \cong \frac{n_2}{n_1}$$

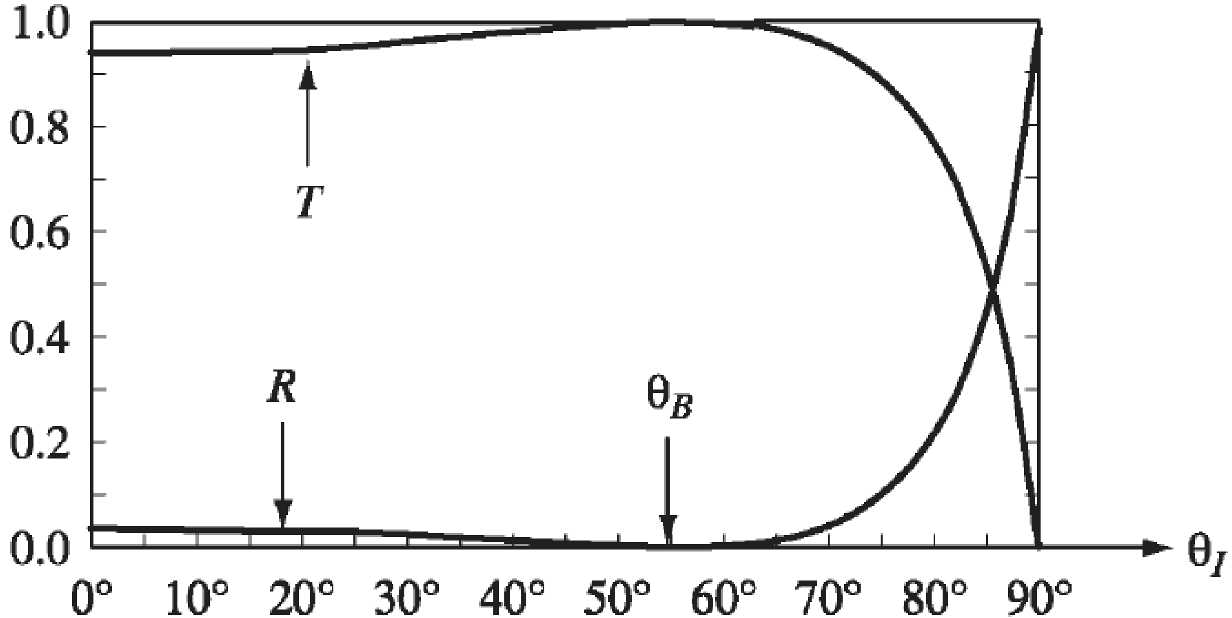


## 9.3.3 Reflection and Transmission at Oblique Incidence

When incident wave is polarized parallel to plane of incidence:

- Intensities:

$$R = \left(\frac{\alpha - \beta}{\alpha + \beta}\right)^2 \ ; \quad T = \frac{4\alpha\beta}{(\alpha + \beta)^2}$$



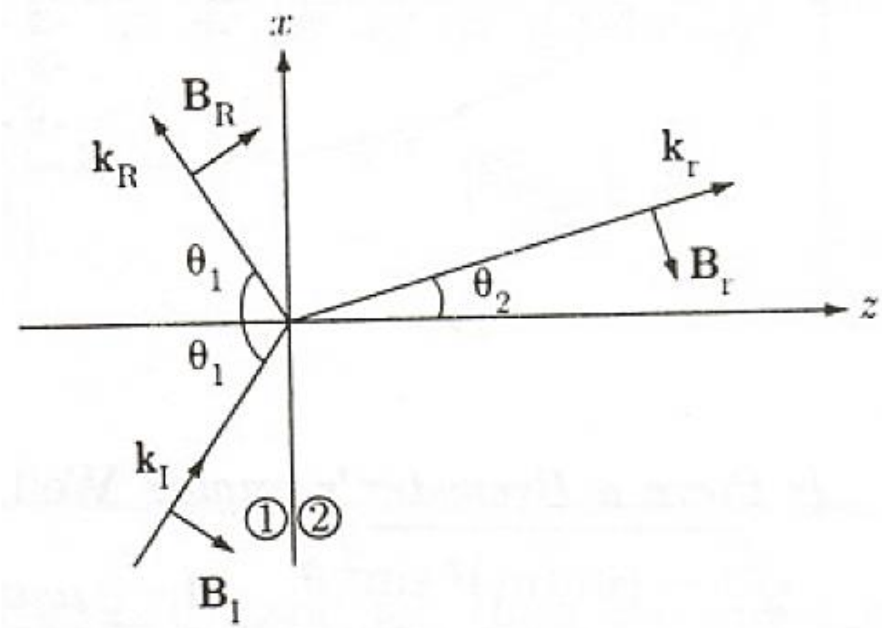
### 9.3.3 Reflection and Transmission at Oblique Incidence

When incident wave is polarized *perpendicular* to plane of incidence:

- Fresnel equations:

$$\tilde{E}_{0R} = \left(\frac{1-\alpha\beta}{1+\alpha\beta}\right)\tilde{E}_{0I}$$

$$\tilde{E}_{0T} = \left(\frac{2}{1+\alpha\beta}\right)\tilde{E}_{0I}$$




- Intensities:

$$R = \left(\frac{1-\alpha\beta}{1+\alpha\beta}\right)^2$$

$$T = \frac{4\alpha\beta}{(1+\alpha\beta)^2}$$

## 9.4 Absorption and Dispersion

### 9.4.1 Electromagnetic Waves in Conductors

The free current $\mathbf{J}_f = \sigma\,\mathbf{E}$ (Ohm's law):

$$\left.\begin{array}{ll} \text{(i)}\ \nabla\cdot\mathbf{E}=0, & \text{(iii)}\ \nabla\times\mathbf{E}=-\dfrac{\partial\mathbf{B}}{\partial t}, \\ \text{(ii)}\ \nabla\cdot\mathbf{B}=0, & \text{(iv)}\ \nabla\times\mathbf{B}=\mu\epsilon\dfrac{\partial\mathbf{E}}{\partial t}+\mu\sigma\mathbf{E}. \end{array}\right\} \quad (9.121)$$

- Combining:

$$\nabla^2\mathbf{E}=\mu\epsilon\frac{\partial^2\mathbf{E}}{\partial t^2}+\mu\sigma\frac{\partial\mathbf{E}}{\partial t}, \quad \nabla^2\mathbf{B}=\mu\epsilon\frac{\partial^2\mathbf{B}}{\partial t^2}+\mu\sigma\frac{\partial\mathbf{B}}{\partial t}$$

- Solution: Transverse waves

$$\tilde{\mathbf{E}}(z,t)=\tilde{\mathbf{E}}_0e^{i(\tilde{k}z-\omega t)}, \quad \tilde{\mathbf{B}}(z,t)=\tilde{\mathbf{B}}_0e^{i(\tilde{k}z-\omega t)}$$

$$\tilde{k}^2=\mu\epsilon\omega^2+i\mu\sigma\omega$$ *Complex wavenumber !*

## 9.4.1 Electromagnetic Waves in Conductors

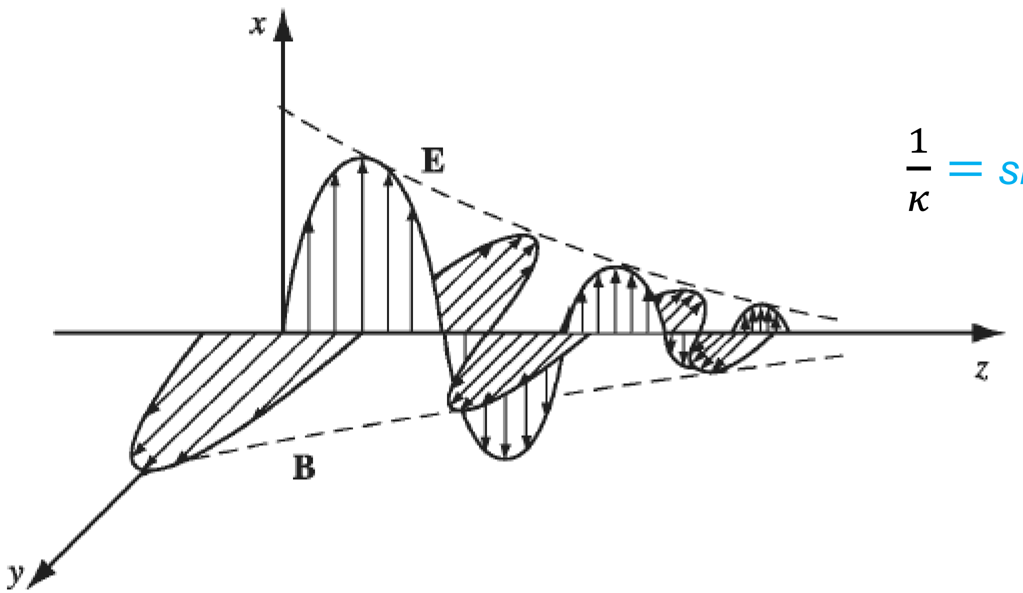
- Complex wavenumber:

$$\tilde{k}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega \quad \Longrightarrow \quad \tilde{k} = k + i\kappa$$

$$\tilde{\mathbf{E}}(z,t) = \tilde{\mathbf{E}}_0 e^{-\kappa z} e^{i(kz-\omega t)}, \quad \tilde{\mathbf{B}}(z,t) = \tilde{\mathbf{B}}_0 e^{-\kappa z} e^{i(kz-\omega t)}$$

- For transverse waves

$$\tilde{\mathbf{E}}(z,t) = \tilde{E}_0 e^{-\kappa z} e^{i(kz-\omega t)}\,\hat{\mathbf{x}} \qquad \tilde{\mathbf{B}}(z,t) = \frac{\tilde{k}}{\omega}\tilde{E}_0 e^{-\kappa z} e^{i(kz-\omega t)}\,\hat{\mathbf{y}}$$




$\frac{1}{\kappa}$ = *skin depth*

### 9.4.1 Electromagnetic Waves in Conductors

- Rewriting the complex wavenumber:

$$\tilde{k} = Ke^{i\phi} \quad \text{where} \quad K = \sqrt{k^2 + \kappa^2} \quad \text{and} \quad \phi = \tan^{-1}(\kappa/k)$$

- Complex amplitudes:

$$\tilde{E}_0 = E_0 e^{i\delta_E} \quad \text{and} \quad \tilde{B}_0 = B_0 e^{i\delta_B} = \frac{Ke^{i\phi}}{\omega} E_0 e^{i\delta_E}$$

$$\delta_B = \delta_E + \phi$$

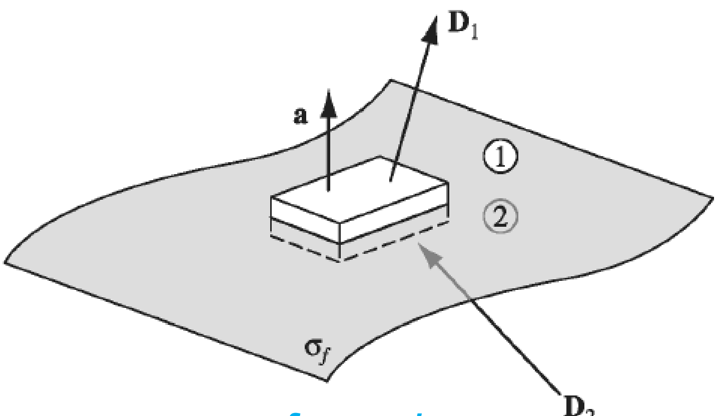
- The real fields:

$$\left.\begin{aligned} \mathbf{E}(z,t) &= E_0 e^{-\kappa z} \cos(kz - \omega t + \delta_E)\,\hat{\mathbf{x}}, \\ \mathbf{B}(z,t) &= B_0 e^{-\kappa z} \cos(kz - \omega t + \delta_E + \phi)\,\hat{\mathbf{y}}. \end{aligned}\right\} \qquad (9.138)$$

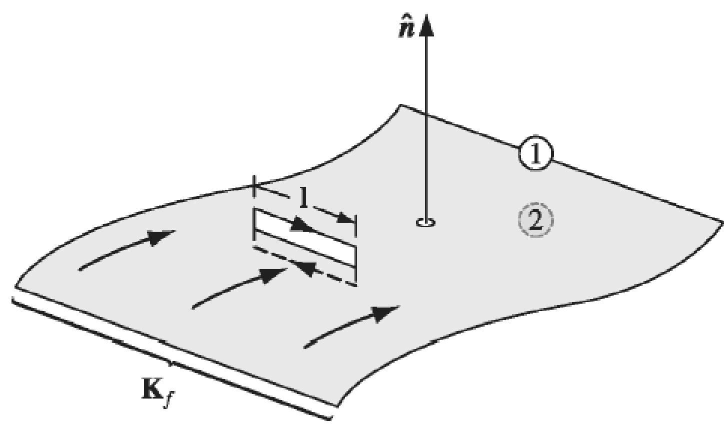
### 9.4.2 Reflection at a Conducting Surface

Boundary conditions:

$$\left.\begin{aligned} &\text{(i)}\ \ \epsilon_1 E_1^{\perp} - \epsilon_2 E_2^{\perp} = \sigma_f, &&\text{(iii)}\ \ \mathbf{E}_1^{\parallel} - \mathbf{E}_2^{\parallel} = \mathbf{0}, \\ &\text{(ii)}\ \ B_1^{\perp} - B_2^{\perp} = 0, &&\text{(iv)}\ \ \frac{1}{\mu_1}\mathbf{B}_1^{\parallel} - \frac{1}{\mu_2}\mathbf{B}_2^{\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}}, \end{aligned}\right\}$$




$\sigma_f$ = *surface charge*




$K_f$ = *surface current*

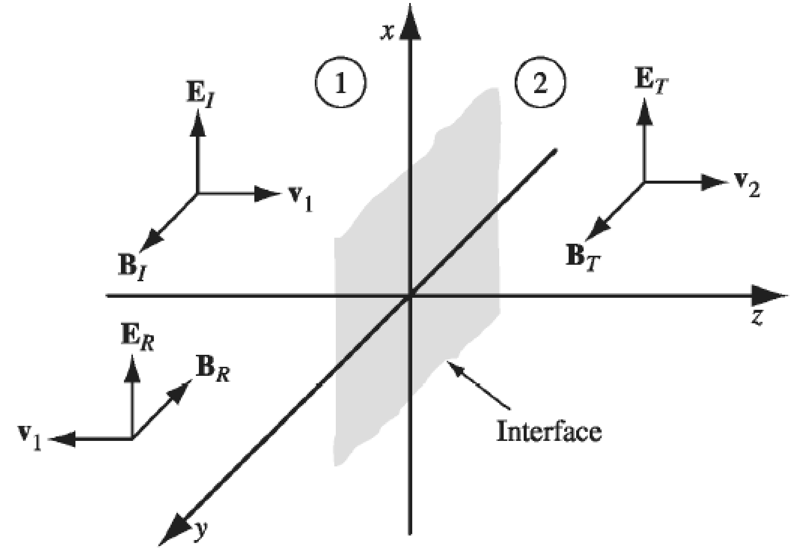
- For Ohmic conductors : $K_f = 0$

## 9.4.2 Reflection at a Conducting Surface

At the plane interface of dielectric (1) and conductor (2):

- Incident wave:

$$\left.\begin{aligned}\tilde{\mathbf{E}}_I(z,t) &= \tilde{E}_{0_I} e^{i(k_1 z-\omega t)}\,\hat{\mathbf{x}},\\ \tilde{\mathbf{B}}_I(z,t) &= \frac{1}{v_1}\tilde{E}_{0_I} e^{i(k_1 z-\omega t)}\,\hat{\mathbf{y}}.\end{aligned}\right\}$$




- Reflected wave:

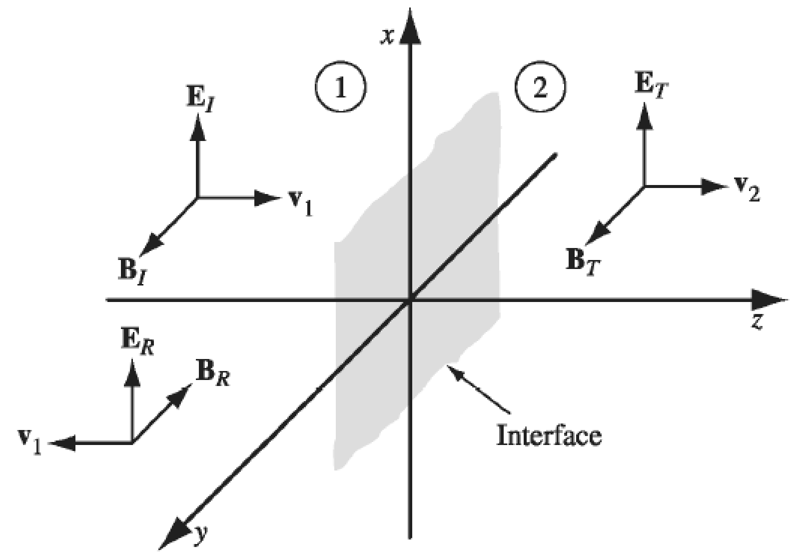
$$\tilde{\mathbf{E}}_R(z,t) = \tilde{E}_{0_R} e^{i(-k_1 z-\omega t)}\,\hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}_R(z,t) = -\frac{1}{v_1}\tilde{E}_{0_R} e^{i(-k_1 z-\omega t)}\,\hat{\mathbf{y}}$$

- Transmitted wave:

$$\tilde{\mathbf{E}}_T(z,t) = \tilde{E}_{0_T} e^{i(\tilde{k}_2 z-\omega t)}\,\hat{\mathbf{x}}, \quad \tilde{\mathbf{B}}_T(z,t) = \frac{\tilde{k}_2}{\omega}\tilde{E}_{0_T} e^{i(\tilde{k}_2 z-\omega t)}\,\hat{\mathbf{y}}$$

### 9.4.2 Reflection at a Conducting Surface

Boundary conditions at interface:




- From (iii): $\mathbf{E}_1^{\parallel} = \mathbf{E}_2^{\parallel}$

$$\tilde{E}_{0I} + \tilde{E}_{0R} = \tilde{E}_{0T}$$

- From (iv): $\mathbf{B}_1^{\parallel}/\mu_1 = \mathbf{B}_2^{\parallel}/\mu_2$

$$\left(\tilde{E}_{0I} - \tilde{E}_{0R}\right)/\mu_1 v_1 = \tilde{k}_2\left(\tilde{E}_{0T}\right)/\mu_2\omega$$

- Solving:

$$\begin{cases} \tilde{E}_{0I} + \tilde{E}_{0R} = \tilde{E}_{0T} \\ \tilde{E}_{0I} - \tilde{E}_{0R} = \tilde{\beta}\tilde{E}_{0T} \end{cases} \Longrightarrow \begin{cases} \tilde{E}_{0R} = \left(\dfrac{1-\tilde{\beta}}{1+\tilde{\beta}}\right)\tilde{E}_{0I} \\ \tilde{E}_{0T} = \left(\dfrac{2}{1+\tilde{\beta}}\right)\tilde{E}_{0I} \end{cases}$$

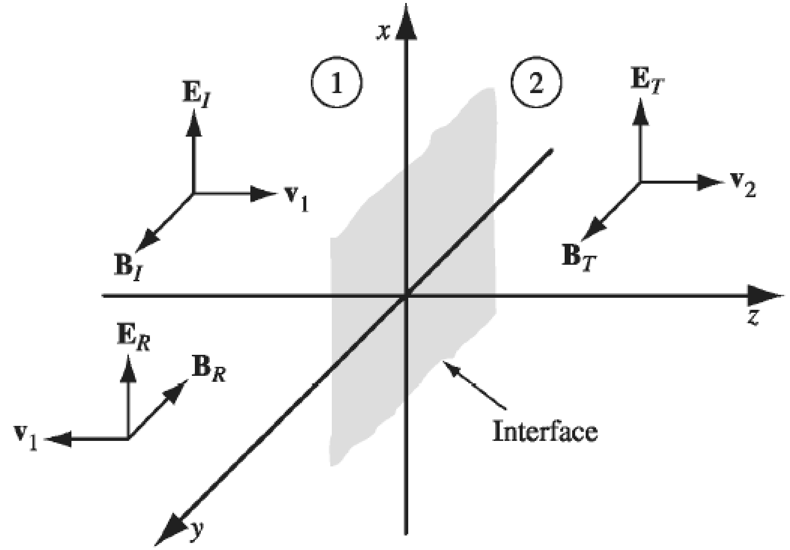
$$\text{where } \tilde{\beta} = \tilde{k}_2\mu_1 v_1/\mu_2\omega$$

## 9.4.2 Reflection at a Conducting Surface

Boundary conditions at interface:

- Solution

$$\begin{cases} \tilde{E}_{0R} = \left(\dfrac{1-\tilde{\beta}}{1+\tilde{\beta}}\right)\tilde{E}_{0I} \\ \tilde{E}_{0T} = \left(\dfrac{2}{1+\tilde{\beta}}\right)\tilde{E}_{0I} \end{cases}$$

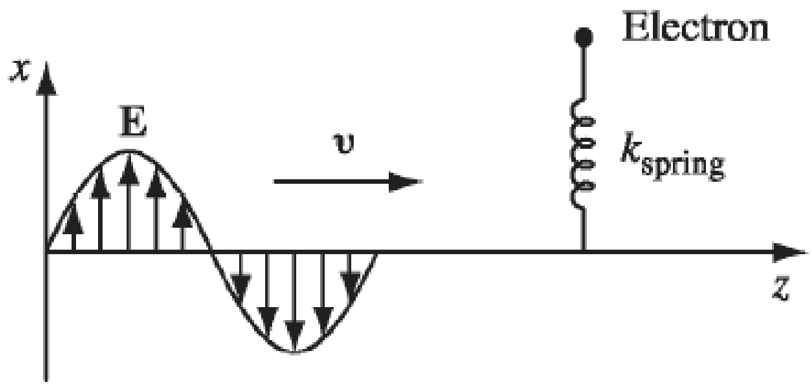



- For perfect conductors ($\sigma = \infty$ ): $\tilde{\beta} = \infty$

$$\tilde{E}_{0T} = 0 \implies \tilde{E}_{0R} = -\tilde{E}_{0I}$$

### 9.4.3 The Frequency Dependence of Permittivity

The electrons in a nonconductor are bound to specific molecules :




- The simplified forces:

$$F_{binding} = -k_{spring}x = -m\omega_0^2 x$$

$$F_{damping} = -m\gamma \frac{dx}{dt}$$

$$F_{driving} = qE = qE_0 \cos(\omega t)$$

### 9.4.3 The Frequency Dependence of Permittivity

- The equation of motion:

$$m\frac{d^2x}{dt^2} + m\gamma\frac{dx}{dt} + m\omega_0^2 x = qE_0\cos(\omega t)$$

$$\boxed{m\frac{d^2\tilde{x}}{dt^2} + m\gamma\frac{d\tilde{x}}{dt} + m\omega_0^2\tilde{x} = qE_0 e^{-i\omega t}}$$

- The solution:

$$\tilde{x} = \tilde{x}_0 e^{-i\omega t} \qquad \text{where} \qquad \tilde{x}_0 = \frac{q/m}{\omega_0^2 - \omega_0^2 - i\gamma\omega}E_0$$

- The dipole moment is the real part of $\tilde{p} = q\tilde{x}(t)$:

$$\tilde{p} = \frac{q^2}{m}\frac{1}{\omega_j^2 - \omega^2 - i\gamma\omega}E_0 e^{-i\omega t}$$

## 9.4.3 The Frequency Dependence of Permittivity

For $N$ molecules per unit volume, each molecule contains $f_j$ electrons with frequency $\omega_j$ and damping $\gamma_j$.

The polarization $\mathbf{P}$ is given by the real part of:

$$\tilde{\mathbf{P}} = \frac{Nq^2}{m}\sum_j \left(\frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega}\right)\tilde{\mathbf{E}} = \varepsilon_0\tilde{\chi}_e\tilde{\mathbf{E}}$$

$$\boxed{\tilde{\chi}_e = \frac{Nq^2}{\varepsilon_0 m}\sum_j \left(\frac{f_j}{\omega_0^2 - \omega^2 - i\gamma_j\omega}\right)}$$

*Complex susceptibility*

- The complex dielectric constant :

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = (1 + \tilde{\chi}_e) = 1 + \frac{Nq^2}{\varepsilon_0 m}\sum_j \left(\frac{f_j}{\omega_0^2 - \omega_j^2 - i\gamma_j\omega}\right)$$

## 9.4.3 The Frequency Dependence of Permittivity

The wave equation for a given frequency :

$$\nabla^2 \tilde{\mathbf{E}} = \tilde{\epsilon}\mu_0 \frac{\partial^2 \tilde{\mathbf{E}}}{\partial t^2}; \qquad \tilde{\mathbf{E}}(z,t) = \tilde{\mathbf{E}}_0 e^{i(\tilde{k}z - \omega t)}$$

- The complex wave number :

$$\tilde{k} = \omega\sqrt{\tilde{\varepsilon}\mu_0} = k + i\kappa$$

- Then :

$$\tilde{\boldsymbol{E}}(z,t) = \tilde{\boldsymbol{E}}_0 e^{-\kappa z} e^{-i(kz-\omega t)}$$

$$\alpha = 2\kappa$$ *Absorption coefficient*

$$n = \frac{ck}{\omega}$$ *Index of refraction*

### 9.4.3 The Frequency Dependence of Permittivity

For gases, the second term of $\varepsilon_r$ is small. We use the approximation:

$$\tilde{k} = \frac{\omega}{c}\sqrt{\tilde{\epsilon}_r} \cong \frac{\omega}{c}\left[1 + \frac{Nq^2}{2m\epsilon_0}\sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j\omega}\right]$$

- Then :

$$\alpha = 2\kappa \cong \frac{Nq^2\omega^2}{m\epsilon_0 c}\sum_j \frac{f_j\gamma_j}{\left(\omega_j^2 - \omega^2\right)^2 + \gamma_j^2\omega^2}$$

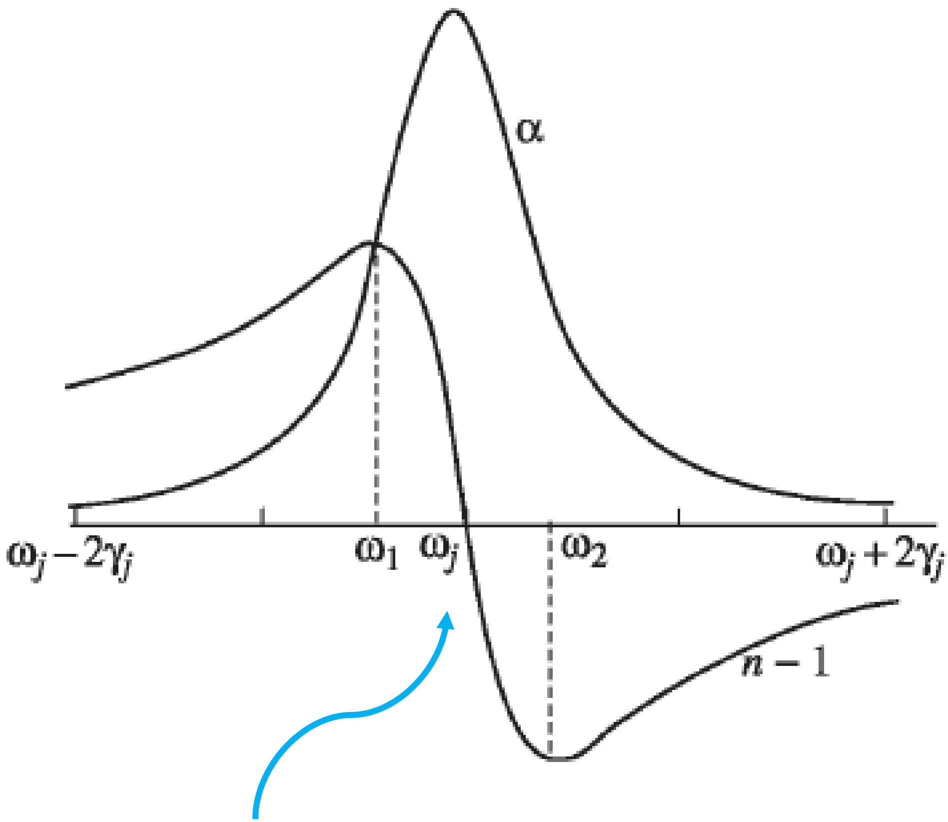
*Absorption coefficient*

$$n = \frac{ck}{\omega} \cong 1 + \frac{Nq^2}{2m\epsilon_0}\sum_j \frac{f_j\left(\omega_j^2 - \omega^2\right)}{\left(\omega_j^2 - \omega^2\right)^2 + \gamma_j^2\omega^2}$$

*Index of refraction*

### 9.4.3 The Frequency Dependence of Permittivity

Plot of $\alpha$ and $n$ near one resonant frequency:




*In the immediate neighborhood of a resonance, the index of refraction drops sharply—called anomalous dispersion.*

### 9.4.3 The Frequency Dependence of Permittivity

Away from the resonances, the damping can be ignored, and the formula for $n$ simplifies:

$$n = 1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j^2 - \omega^2}$$

- For transparent materials, the nearest significant resonances typically lie in the ultraviolet, so that $\omega < \omega_j$ :

$$n = 1 + \left( \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j^2} \right) + \omega^2 \left( \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j^4} \right)$$

- Then (using $\lambda = 2\pi c/\omega$) :

$$n = 1 + A\left(1 + \frac{B}{\lambda^2}\right)$$

*Dispersion equation (Cauchy's formula)*

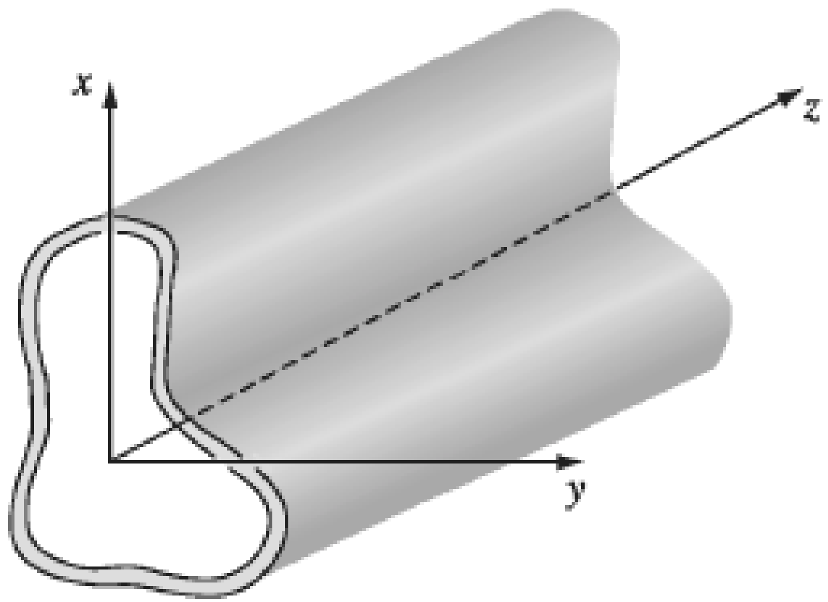
## 9.5 Waves in One Dimension

### 9.5.1 Wave Guides

Consider waves confined to the interior to some sort of pipe, or ***wave guide***.

- The overall geometry of the pipe should be large compared with the radiation wavelength.
- The pipe is made of a perfect conductor so $\mathbf{E} = 0$ and $\mathbf{B} = 0$ within the material
- The boundary conditions at the inner wall are

$$\mathbf{E}^{\parallel} = 0\,; \quad \mathbf{B}^{\perp} = 0$$



### 9.5.1 Wave Guides

Monochromatic plane wave solutions:

$$\left.\begin{aligned} &\text{(i)}\quad \tilde{\mathbf{E}}(x, y, z, t) = \tilde{\mathbf{E}}_0(x, y)e^{i(kz-\omega t)}, \\ &\text{(ii)}\quad \tilde{\mathbf{B}}(x, y, z, t) = \tilde{\mathbf{B}}_0(x, y)e^{i(kz-\omega t)}. \end{aligned}\right\}$$

- Must satisfy Maxwell's equations and boundary conditions:

$$\left.\begin{aligned} &\text{(i)}\quad \nabla \cdot \mathbf{E} = 0, \qquad \text{(iii)}\quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ &\text{(ii)}\quad \nabla \cdot \mathbf{B} = 0, \qquad \text{(iv)}\quad \nabla \times \mathbf{B} = \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}. \end{aligned}\right\}$$

- It turns out that within a waveguide we are not necessarily limited to transverse solutions only. We include the longitudinal components of the fields when plugging in the coordinates into our boundary conditions and generic solutions

$$\tilde{\mathbf{E}}_0 = E_x\,\hat{\mathbf{x}} + E_y\,\hat{\mathbf{y}} + E_z\,\hat{\mathbf{z}}, \qquad \tilde{\mathbf{B}}_0 = B_x\,\hat{\mathbf{x}} + B_y\,\hat{\mathbf{y}} + B_z\,\hat{\mathbf{z}}, \qquad (9.178)$$

### 9.5.1 Wave Guides

- From Maxwell's equations:

$$\left.\begin{array}{llll}
\text{(i)} & \dfrac{\partial E_y}{\partial x}-\dfrac{\partial E_x}{\partial y}=i\omega B_z, & \text{(iv)} & \dfrac{\partial B_y}{\partial x}-\dfrac{\partial B_x}{\partial y}=-\dfrac{i\omega}{c^2}E_z, \\
\text{(ii)} & \dfrac{\partial E_z}{\partial y}-ikE_y=i\omega B_x, & \text{(v)} & \dfrac{\partial B_z}{\partial y}-ikB_y=-\dfrac{i\omega}{c^2}E_x, \\
\text{(iii)} & ikE_x-\dfrac{\partial E_z}{\partial x}=i\omega B_y, & \text{(vi)} & ikB_x-\dfrac{\partial B_z}{\partial x}=-\dfrac{i\omega}{c^2}E_y.
\end{array}\right\}$$

- Solution:

$$\left.\begin{array}{ll}
\text{(i)} & E_x=\dfrac{i}{(\omega/c)^2-k^2}\left(k\dfrac{\partial E_z}{\partial x}+\omega\dfrac{\partial B_z}{\partial y}\right), \\
\text{(ii)} & E_y=\dfrac{i}{(\omega/c)^2-k^2}\left(k\dfrac{\partial E_z}{\partial y}-\omega\dfrac{\partial B_z}{\partial x}\right), \\
\text{(iii)} & B_x=\dfrac{i}{(\omega/c)^2-k^2}\left(k\dfrac{\partial B_z}{\partial x}-\dfrac{\omega}{c^2}\dfrac{\partial E_z}{\partial y}\right), \\
\text{(iv)} & B_y=\dfrac{i}{(\omega/c)^2-k^2}\left(k\dfrac{\partial B_z}{\partial y}+\dfrac{\omega}{c^2}\dfrac{\partial E_z}{\partial x}\right).
\end{array}\right\} \qquad (9.180)$$

### 9.5.1 Wave Guides

- Uncoupled equations:

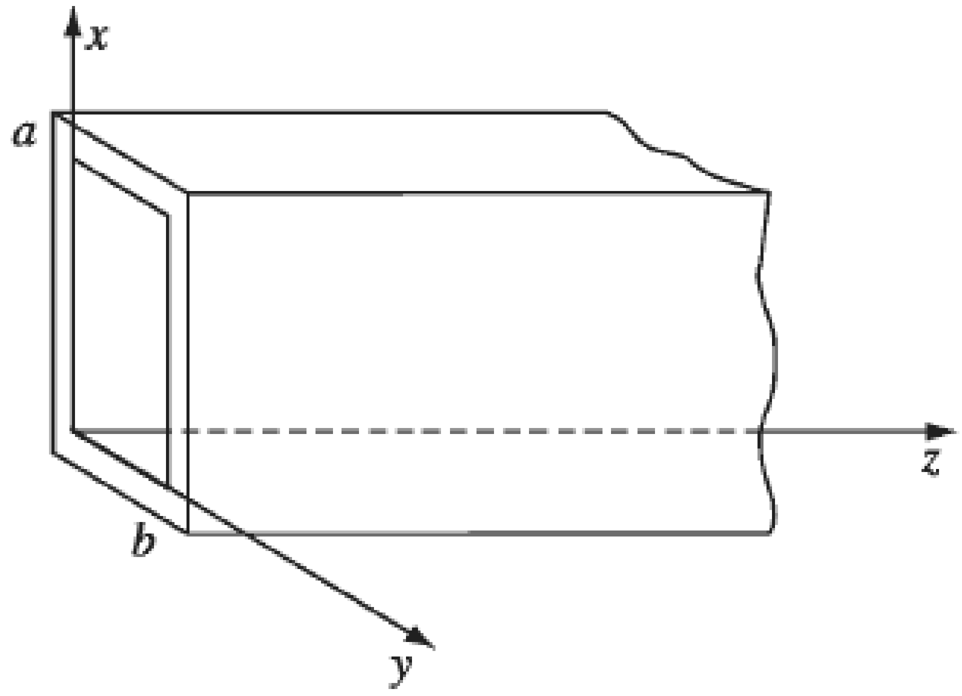
$$\left.\begin{array}{ll} \text{(i)} & \left[\dfrac{\partial^2}{\partial x^2}+\dfrac{\partial^2}{\partial y^2}+(\omega/c)^2-k^2\right]E_z=0, \\ \text{(ii)} & \left[\dfrac{\partial^2}{\partial x^2}+\dfrac{\partial^2}{\partial y^2}+(\omega/c)^2-k^2\right]B_z=0. \end{array}\right\} \qquad (9.181)$$

- For trivial solutions, we can come up with separate classes of solutions:
  - If $E_z = 0$, we call the solutions **transverse electric (TE)** waves.
  - If $B_z = 0$, they are called **transverse magnetic (TM)** waves.
  - If both are zero, we call them **TEM** waves (it turns out that TEM waves can't occur in a hollow wave guide)

### 9.5.2 Rectangular Wave Guide

Consider a rectangular wave guide with height *a* and width *b*. Let us specifically take a look at the **TE** ($E_z = 0$) solutions (the process is very similar for **TM** waves, except that the boundary conditions are flipped).

$$\left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + (\omega/c)^2 - k^2\right] B_z = 0. \qquad (9.181\text{ii})$$



## 9.5.2 Rectangular Wave Guide

- Start with the method of separation of variables.

$$B_z(x, y) = X(x)Y(y)$$

- Substitution:

$$Y\frac{d^2X}{dx^2} + X\frac{d^2Y}{dy^2} + \left[(\omega/c)^2 - k^2\right]XY = 0$$

$$\text{(i)}\quad \frac{1}{X}\frac{d^2X}{dx^2} = -k_x^2, \quad \text{(ii)}\quad \frac{1}{Y}\frac{d^2Y}{dy^2} = -k_y^2$$

- General solution:

$$X(x) = C_1 \sin(k_x x) + C_2 \cos(k_x x)$$

$$Y(y) = D_1 \sin\left(k_y y\right) + D_2 \cos(k_y y)$$

with

$$-k_x^2 - k_y^2 + (\omega/c)^2 - k^2 = 0 \qquad (9.183)$$

## 9.5.2 Rectangular Wave Guide

- Applying boundary conditions $B_x = 0$ (from $\mathbf{B}^{\perp} = 0$) :

$$\left.\frac{\partial X}{\partial x}\right)_{x=0} = 0; \qquad \left.\frac{\partial X}{\partial x}\right)_{x=a} = 0$$

- Substitution:

$$X(x) = C_2 \cos(k_x x)$$

$$k_x = \frac{m\pi}{a} \qquad (m = 0,1,2,\dots)$$

- The same goes for $Y$ :

$$Y(y) = D_2 \cos(k_y y)$$

$$k_y = \frac{n\pi}{b} \qquad (n = 0,1,2,\dots)$$

## 9.5.2 Rectangular Wave Guide

The $\mathrm{TE}_{mn}$ mode :

$$B_z(x, y) = B_0 \cos(k_m x) \cos(k_n x)$$

- The wavenumber:

$$k = \sqrt{(\omega/c)^2 - \pi^2[(m/a)^2 + (n/b)^2]}$$
$$= \frac{1}{c}\sqrt{\omega^2 - {\omega_{mn}}^2}$$

- The cut-off frequency :

$$\omega_{mn} = c\pi\sqrt{\,[(m/a)^2 + (n/b)^2]}$$

## 9.5.2 Rectangular Wave Guide

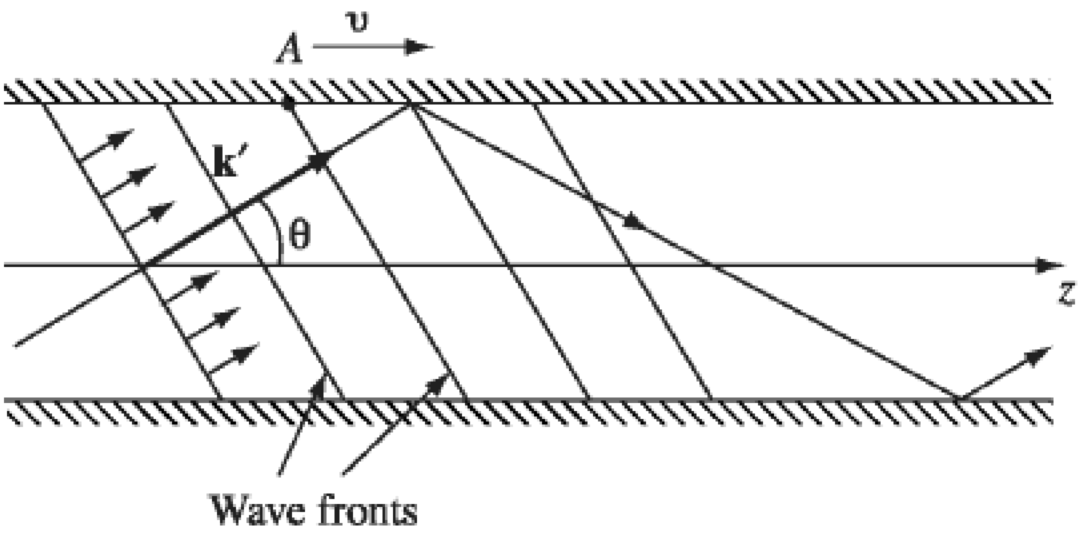
Visualization :

- Consider an ordinary *plane* wave, traveling at an angle θ to the *z* axis, and reflecting perfectly off each conducting surface

- The wavenumber:

$$\omega = c\sqrt{(k)^2 + [(m\pi/a)^2 + (n\pi/b)^2]} \equiv c|\mathbf{k}'|$$

$$\mathbf{k}' = \frac{m\pi}{a}\hat{\boldsymbol{x}} + \frac{n\pi}{b}\hat{\boldsymbol{y}} + k\,\hat{\mathbf{z}} = k_x\hat{\boldsymbol{x}} + k_y\hat{\boldsymbol{y}} + k_z\hat{\mathbf{z}}$$



## 9.5.2 Rectangular Wave Guide

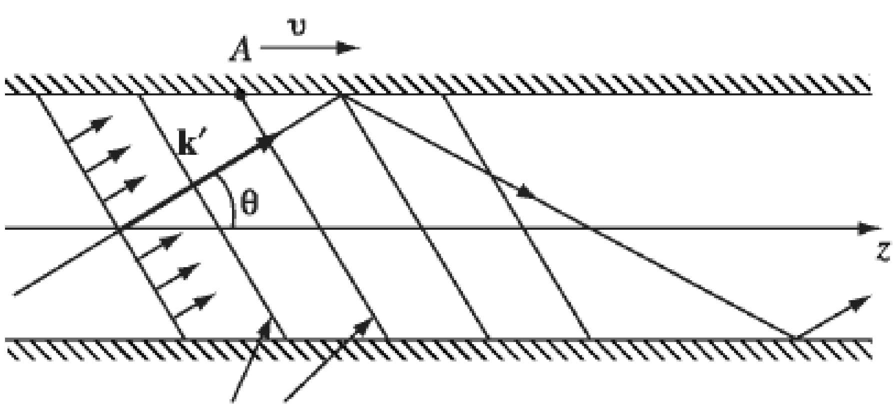
Visualization :

- The angle $\theta$ satisfies the condition

$$\cos\boldsymbol{\theta} = \frac{k}{|\mathbf{k}'|} = \sqrt{1 - (\omega_{mn}/\omega)^2}$$

- The plane wave travels at speed *c*, but because it is going at an angle $\theta$ to the *z* axis, its net velocity down the wave guide is

$$v_g = c \cdot \cos\theta = c\sqrt{1 - (\omega_{mn}/\omega)^2}$$

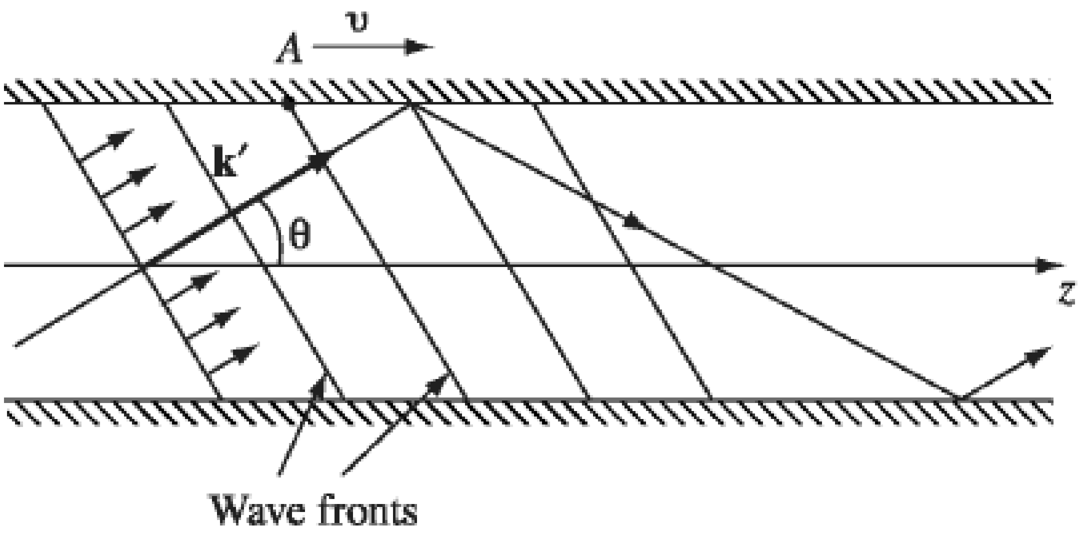


## 9.5.2 Rectangular Wave Guide

Visualization :

- The *wave* velocity, on the other hand, is the speed of the wave fronts down the pipe

$$v = \frac{c}{\cos\theta} = \frac{c}{\sqrt{1-(\omega_{mn}/\omega)^2}}$$



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